Hyperspectral Real-Time Local Anomaly Detection Based on Finite Markov via Line-by-Line Processing

Shihui Liu, Meiping Song, *Member, IEEE*, Bing Xue, *Fellow, IEEE*, Chein-I Chang, *Life Fellow, IEEE*, and Mengjie Zhang, *Fellow, IEEE*

***Abstract*— Real-time anomaly detection technique can efficiently and effectively leverage available data and operates in tandem with data collection, avoiding dependence on unacquired spectral data. Nonetheless, there are no restrictions or discussions regarding the scope of utilization for existing data. Overloading the analysis with excessive information, particularly encompassing dynamically changing background scenes, can introduce interference, undermining the statistical characteristics of the data and hampering anomaly detection. Studies indicate that local anomaly detection can enhance detection performance. Consequently, determining the optimal scope of the row space within the context of real-time line-by-line processing by integrating local processing and real-time technology stands pivotal in enhancing efficacy. In order to realize real-time hyperspectral local anomaly detection, based on the most widely used push-broom hyperspectral imaging sensor, this article proposes a finite Markov local real-time correlation matrix R anomaly detection (FMLRT-RAD) by studying the similarity of spectra in adjacent regions of the same substance and the independence of spectra in different regions in hyperspectral images. FMLRT-RAD can adaptively determine the size of the local background region, and solve the challenging task of selecting a suitable data range for local background suppression when an imaging sensor obtains a large amount of data. In addition, based on sample correlation matrix R anomaly detection, two different correlation matrix representations are designed for dynamically updating finite local samples. Woodbury matrix identity is used to update background suppression, and corresponding update equations with causal recursion characteristics are derived to achieve local real-time anomaly detection to further reduce time consumption and improve detection capability. The experimental results of several real-world hyperspectral image datasets show that the detector has superior detection performance compared with other advanced detectors.**

***Index Terms*— Finite Markov, hyperspectral images, push-broom hyperspectral imaging sensor, real-time local anomaly detection, Woodbury matrix identity.**

I. Introduction

HYPERSPECTRAL anomaly detection is a key problem in hyperspectral image analysis [1], [2], [3], [4]. Hyperspectral anomaly detection is to identify the anomaly target from the background according to the spectral characteristics difference between the target and the image background without prior knowledge of the target spectrum [5], [6]. Due to the advantages of not requiring any prior knowledge, hyperspectral anomaly detection has high practical application value in environmental monitoring, mineral exploration, and food quality detection.

In the past few decades, many hyperspectral anomaly detection methods have been proposed [7], [8], [9]. For example, the Reed–Xiaoli detector (RXD) proposed by Reed and Yu [10] is widely regarded as the benchmark algorithm in this field. It assumes that the probability density function of the background is in line with the multivariate Gaussian distribution, and does not require the prior information of the target spectrum, but only needs the global mean and background covariance matrix to evaluate the anomaly pixels. Later, Chang and Chiang [11] proposed correlation matrix R-based anomaly detection (R-AD), which simplifies the RXD operation without affecting the detection effectiveness and has become one of the more widely used anomaly detection algorithms. Based on super resolution (SR), Li and Zhang [12] proposed a hyperspectral anomaly detection algorithm, which reflected spectral anomaly changes through an extended linear mixture model. The image SR was used to inject the ground object information of the mixture model and the spatial correlation was combined with the global Reed–Xiaoli method and ground object correlation detection for anomaly detection.

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Shihui Liu is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, and also with the Center for Data Science and Artificial Intelligence and the School of Engineering and Computer Science, Victoria University of Wellington, Wellington 6140, New Zealand (e-mail: 1473245950@qq.com).

Meiping Song is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China (e-mail: smping@163.com).

Bing Xue and Mengjie Zhang are with the Center for Data Science and Artificial Intelligence and the School of Engineering and Computer Science, Victoria University of Wellington, Wellington 6140, New Zealand (e-mail: bing.xue@ecs.vuw.ac.nz; mengjie.zhang@ecs.vuw.ac.nz).

Chein-I Chang is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, and also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore, MD 21250 USA (e-mail: cchang@umbc.edu).

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Based on the visual attention model, Xiang et al. [13] proposed a hyperspectral anomaly detection implementation strategy with adaptive weight background subtraction visual attention and background subtraction (VABS), total variation curvature filter is introduced into hyperspectral anomaly detection for background subtraction. Based on low-rank and sparse matrix decomposition, Ma et al. [14] proposed a feature extraction and background purification anomaly detector (FEBPAD) that combines fractional Fourier transform with low-rank sparse matrix factorization fractional Fourier transform (LRaSMD). Based on autoencoder (AE), Fan et al. [15] proposed a robust graph acoustic emission (RGAE) detection method by embedding a superpixel segmentation-based graph regularization term into the robust AE framework with $\ell_{2,1}$-norm. Based on the idea of transforming the target detector into the corresponding anomaly detection, Chang [16] designed the target-to-anomaly conversion-derived anomaly detector (TAC-AD), and proposed an effective anomaly space (EAS) method combined with TAC-AD to keep anomalies in EAS and remove interference and background noise (BKG). Based on low-rank representation, Guo et al. [17] proposed an antinoise hierarchical mutual-incoherence-induced discriminative learning (AHMID) method for hyperspectral anomaly detection, which utilizes structural incoherence constraints to separate background and anomalies. Based on deep learning, Xiao et al. [18] proposed a hyperspectral anomaly detection method based on a spatial-spectral dual-window mask transformer, which fully extracts features from global and local perspectives, and adaptively suppresses the reconstruction of anomaly targets. It can be observed that the above-mentioned method, utilizing statistics of background pixels across the entire hyperspectral image is considered a global approach and generally performs well in most situations. However, due to the presence of anomalies, the accuracy of background modeling and the complexity of computation make it not suitable for scenarios with weak anomalies or complex backgrounds, limiting the performance of global methods.

In contrast, local anomaly detection using small neighborhood statistics has recently been extensively studied. For example, Molero et al. [19] proposed a local RX detector, which uses a sliding local window and only considers a set of neighboring pixels around the pixel to be detected. Instead of having a single window to estimate local background statistics, Tan et al. [20] proposed a local summation anomaly detection based on collaborative representation and inverse distance weight (LSAD-CR-IDW) adopts the local summation dual-window sliding strategy and using the pixels collected between the dual-windows to represent the local background information of the pixel to be measured. Based on collaborative representation detection (CRD), Hou et al. [21] proposed a collaborative representation detector with background purification and saliency weight method, a saliency weight is introduced into the collaborative representation, and the saliency weight of the pixel to be measured is estimated from the local background pixels in the inner window. Similarly, based on CRD, Imani [22] proposed a collaborative representation based on the adaptive window CR, which uses the results of two segmentation maps with different numbers of superpixels to find inner and outer windows of appropriate size for each test pixel, but it has the problem of too long running time. Although local anomaly detection algorithms have demonstrated good performance on tested hyperspectral data, the sliding of the neighborhood window directly impacts the detection results. The manually set window size parameter cannot meet the requirement of different datasets requiring different values. Therefore, determining the window size adaptively based on the hyperspectral image is a challenging issue yet to be resolved in local anomaly detection.

Nevertheless, hyperspectral data processing usually has high time complexity, which limits its application in certain industries that require high timeliness, such as environmental monitoring, disaster site detection, industrial production detection, and other practical scenarios, where data should be processed immediately after acquisition. In other words, only the previous data and the data that was just fetched can be used, and no unobtained data samples can be used. It is worth noting that the push-broom hyperspectral imaging sensor is one of the widely used imaging spectrometers, which makes the image data perceived line-by-line [23]. In recent years, many real-time anomaly detection algorithms have been proposed in the field of hyperspectral anomaly detection to find anomalies in real-time [24]. For example, the concept of causal processing was first introduced into hyperspectral real-time anomaly detection, Chen et al. [25] proposed two line-by-line real-time causal processing versions (RT-CR-RXD and RT-CK-RXD) based on RXD of the sample covariance matrix K and correlation matrix R, respectively. Based on Cholesky's decomposition, Zhao and Xi-Feng [26] proposed a pixel-wise local real-time kernel RX detector (LRT-KRXD-CD), which satisfies the causal relationship in real-time systems through a local causal sliding array window. In the same year, Zhao et al. [27] proposed a recursive local summation RX anomaly detection method based on a sliding window, which selects the optimal size of a real-time sliding window by detection performance and sums multiple related local background statistics. Besides RXD-based, there are other categories. For example, based on the method of orthogonal subspace projections (OSPs), Díaz et al. [28] proposed a line-by-line fast hyperspectral image anomaly detector (LbL-FAD), which identifies anomalies pixels based on the automatic threshold estimation of the background distribution while processing each pixel. Based on the sparse representation, Chen et al. [29] proposed a fast random projection low-rank approximation sparse representation method, which only iteratively decomposes the sparse matrix to the zero matrices, and then adds all the decomposed sparse matrices to improve the detectability and detection speed of anomalies in the sparse matrix. Based on the CRD, Lin and Lin [30] proposed a real-time hyperspectral detector with collaborative superpixel representation with anomaly boundary refinement, which uses superpixel segmentation to cluster images and no longer detects pixel-by-pixel, but performs collaborative representation on each superpixel. It can be seen from the above methods that hyperspectral real-time local anomaly detection technique has not been well explored and most real-time local anomaly detection operators are designed

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with window models. However, it can avoid the detection failure problem caused by the global method because the anomaly target only exists in the local scope and is submerged in the global scope. Traditionally, the local window size is manual design, however, is time-consuming, and conducts a large number of experiments. When getting new data, the window size may not be effective. The local anomaly detection window size task is very challenging due to the high image variation and the risk of poor generalization. Therefore, this article focuses on automatically determining the window size based on the given data in an effective and efficient way for real-time processing in hyperspectral local anomaly detection.

In this study, different calculation methods should be used in different ways. Two correlation matrix representations are designed for dynamically updated finite local samples, and the corresponding update equations with causal recursive characteristics are derived to realize real-time local anomaly detection, significantly reducing the computational complexity and data storage space, and improving the detection performance. The major contributions of this work are summarized as follows.

1) Based on the finite Markov, a method is proposed to determine the starting and ending lines of the same substance based on the data samples obtained by the push-broom hyperspectral imaging sensor. In other words, the appropriate data range is determined for local anomaly detection to implement background suppression. This significantly differs from the existing hyperspectral local anomaly detection methods that select the window size by experimental performance. Notably, anomaly detection has no prior spectral information available about the background or anomalies, and guides local real-time anomaly detection by confirming the line where the anomaly is located during the line-by-line acquisition of the data, enabling users to more clearly identify the anomaly location.
2) Based on the line-by-line data acquisition format of the imaging sensor and the judgment of anomaly lines, a new data utilization method is proposed. This includes two ways to use the data: first, continuously accumulating newly acquired data, suitable for situations where the current line has no anomaly. Second, simultaneously removing previously collected data lines from the background while obtaining new data lines, suitable for situations where the current line has an anomaly, avoiding the impact of redundant data and ensuring the stability of the detection results.
3) Based on the R-AD, a local real-time anomaly detection operator is derived using Woodbury matrix identity. This operator replaces the matrix inversion of all data samples, thus speeding up the detection process and reducing the complexity of the operation. In addition, the calculation performance is not affected in the process of discarding the old background information, and the detection accuracy is improved. Users can clearly see how the anomaly detector performs different levels of background suppression.

The rest of this article is organized as follows. In Section II, the hyperspectral real-time local anomaly detection algorithm based on finite Markov line-by-line processing is proposed. Section III gives the corresponding computation and time complexity analysis. In Section IV, the experiment design is introduced. Section V gives the experimental results on five datasets to demonstrate the performance of the proposed algorithm. Section VI concludes this article.

II. Methodology

In this section, we first introduce the proposed finite Markov local real-time correlation matrix R anomaly detection (FMLRT-RAD) method and the main components of the proposed method, as well as provide the FMLRT-RAD algorithm details.

A. Information Content of Hyperspectral Images

In order to identify anomaly lines in the acquired hyperspectral image data, a method to calculate the information content of the data line-by-line is established. Hyperspectral images are stored as digital images [31]. The digital image is a data-form image obtained after sampling and quantization of optical images or analog signals reflected by ground objects. It is worth noting that due to the existence of texture, the gray levels among pixels in hyperspectral images are correlated, the correlation is affected by a certain region. According to information theory [32], [33], the digital image belongs to the Markov signal source which emits variable sequences in discrete sources. Therefore, for a source satisfying the Markov property, that is, the variable emitted by the source is only related to the previous m variable, but not to the variable appearing before it. Suppose to I be a long sequence output by the source, then the sequence entropy of the source is

$$\begin{aligned} H(X) &= H(X_1X_2,\ldots,X_I) \\ &= H(X_1)+H(X_2/X_1)+\cdots+H(X_I/X_1X_2,\ldots,X_{I-1}). \end{aligned} \tag{1}$$

Its probability can be expressed as

$$P(x_t/x_{t-1},x_{t-2},\ldots,x_{t-m},\ldots)=P(x_t/x_{t-1},x_{t-2},\ldots,x_{t-m}). \tag{2}$$

Its limit entropy is converted to

$$\begin{aligned} H_\infty(X) &= \lim_{I\to\infty} H(X_I/X_1X_2,\ldots,X_{I-1}) \\ &= H(X_{m+1}/X_1X_2,\ldots,X_m)=H_{m+1}(X). \end{aligned} \tag{3}$$

If the variables are independent, $H(X_n/X_{n-1})=H(X_n)$. Then (1) becomes

$$\begin{aligned} H(X) &= H(X_1)+H(X_2)+\cdots+H(X_{I-1}) \\ &= -p_{x_1}\log p_{x_1}-p_{x_2}\log p_{x_2}-\cdots-p_{x_{i-1}}\log p_{x_{i-1}} \\ &= -\sum_i p(x_i)\log p(x_i). \end{aligned} \tag{4}$$

After determining the correlation range of the hyperspectral image $\mathbf{X}$, the information of the whole image can be accurately

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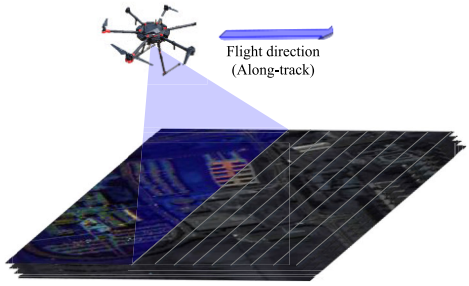


Fig. 1. Schematic of the principle of line-by-line data acquisition by a hyperspectral imaging sensor.

calculated by using a discrete Markov source. Assume that the source is

$$\mathbf{X} = \begin{bmatrix} x_{0,0} & x_{0,1} & \cdots & x_{0,n-1} \\ x_{1,0} & x_{1,1} & \cdots & x_{1,n-1} \\ \vdots & \vdots & \vdots & \vdots \\ x_{m-1,0} & x_{m-1,1} & \cdots & x_{m-1,n-1} \end{bmatrix} \tag{5}$$

where the variable $x_{i,j} \in (g_0, g_1, \ldots g_m) = (0, 1, \ldots 255)$, $x_{i,j}$ is a 256 grayscale images, assuming g_1 is related to only t gray levels and not to the others, the hyperspectral image $\mathbf{X}$ entropy is

$$\begin{aligned} H(\mathbf{X}) &= H(\mathbf{X}_i/\mathbf{X}_{i-1}\mathbf{X}_{i-2}, \ldots, \mathbf{X}_{i-t}) \\ &= -\sum_{i=0}^{m}\sum_{j=0}^{t} p\left(g_i/g_{i-j}\right) \log p\left(g_i/g_{i-j}\right)(i-j \geq 0) \end{aligned} \tag{6}$$

where $p(g_i/g_{i-j}) = p(g_i/g_{i-1}g_{i-2}\ldots g_{i-j})(i-j \geq 0)$.

B. Finite Markov Model for the Amount of Interline Information in Hyperspectral Images

The data acquisition format of hyperspectral imaging sensors has an important impact on the real-time processing of data. At present, four data acquisition formats have been widely used by hyperspectral imaging sensors, namely band-interleaved-by-pixel/sample (BIP/BIS) format, band-interleaved-byline (BIL) format, band sequential (BSQ) format, and multiline multiband (MLMB) format [34]. Among them, the BIP/BIS format collects the full band information data pixel-by-pixel, while the BSQ format collects the data information band-by-band. In MLMB format, the imaging system of a multiarray filter spectral imaging camera is used to obtain multiple lines and multiple bands. Among various data acquisition formats, the BIL format can capture full band information line-by-line, as shown in Fig. 1. There is no doubt that in the field of real-time processing, the BIL format is more suitable than others because push-broom imagery has become the main way widely adopted in the field of hyperspectral data acquisition. Therefore, the design of this article will revolve around the BIL format.

Assume that $\mathbf{X}_i = (\mathbf{x}_1^i(1), \ldots, \mathbf{x}_1^i(B), \mathbf{x}_2^i(1), \ldots, \mathbf{x}_n^i(B))^T$ is all the pixel data of B bands in line i of hyperspectral image $\mathbf{X}$, n is the total number of pixels in a line of hyperspectral data, and x_B^i is the data of B bands of a specific pixel in a line i of hyperspectral image $\mathbf{X}$. If $\mathbf{X}_{Bn}^i$ is the 2-D data of B bands in line with the hyperspectral image $\mathbf{X}$, then the size of $\mathbf{X}_{Bn}^i$ data is $B*n$ and $\mathbf{X}_{Bn}^i = (\mathbf{x}_1^i, \mathbf{x}_2^i, \ldots, \mathbf{x}_n^i)^T$, whose specific expression is shown in the following:

$$\begin{aligned} \mathbf{X}_{Bn}^i &= \left(\mathbf{x}_1^i, \mathbf{x}_2^i, \ldots, \mathbf{x}_n^i\right)^T \\ &= \begin{bmatrix} x_{11}^i & \cdots & x_{1(B-1)}^i & x_{1B}^i \\ \vdots & & \vdots & \vdots \\ x_{n1}^i & \cdots & x_{n(B-1)}^i & x_{nB}^i \end{bmatrix}. \end{aligned} \tag{7}$$

$\mathbf{R}_{BB}^i$ is the correlation matrix of B bands in line i of the hyperspectral image $\mathbf{X}_{Bn}^i$, then $\mathbf{R}_{BB}^i$ data size is $B*B$ and $\mathbf{X}_{Bn}^i = (\mathbf{x}_1^i, \mathbf{x}_2^i, \ldots, \mathbf{x}_n^i)^T$, whose specific expression is shown in the following:

$$\begin{aligned} \mathbf{R}_{BB}^i &= \frac{1}{n}\left(\mathbf{x}_1^i, \mathbf{x}_2^i, \ldots, \mathbf{x}_n^i\right)^T \left(\mathbf{x}_1^i, \mathbf{x}_2^i, \ldots, \mathbf{x}_n^i\right) \\ &= \frac{1}{n}\begin{bmatrix} x_{11}^i & \cdots & x_{1(B-1)}^i & x_{1B}^i \\ \vdots & & \vdots & \vdots \\ x_{n1}^i & \cdots & x_{n(B-1)}^i & x_{nB}^i \end{bmatrix} \\ &\quad \times \begin{bmatrix} x_{11}^i & \cdots & x_{n1}^i \\ \vdots & & \vdots \\ x_{1(B-1)}^i & x_{n(B-1)}^i & \\ x_{1B}^i & \cdots & x_{nB}^i \end{bmatrix}. \end{aligned} \tag{8}$$

Then, by calculating the trace of the correlation matrix $\mathbf{R}_{BB}^i$, the information amount of the current line i can be obtained, and its specific expression is shown in the following:

$$H(\mathbf{X}_i) = \mathrm{tr}\left(\mathbf{R}_{BB}^i\right) = \sum_{i=1}^{B} x_{ii} = x_{11} + x_{22} + \cdots + x_{BB}. \tag{9}$$

The amount of line information of hyperspectral images is often calculated by converting the line-by-line acquired hyperspectral image $\mathbf{X}_i$ into a 2-D matrix $\mathbf{X}_{Bn}^i$, and then obtaining the correlation matrix $\mathbf{R}_{BB}^i$ by $\mathbf{X}_{Bn}^i$ conversion. The whole data conversion principle is shown in Fig. 2. The amount of information in different lines of hyperspectral images can be calculated by the above method. For example, the amount of information in lines of the Pavia Centre dataset is shown in Fig. 3, where Fig. 3(a) is the false color image of the Pavia Centre dataset, Fig. 3(b) is the ground truth and Fig. 3(c) is the amount of information in different lines. The Pavia Centre dataset will be introduced in detail in the experiment design. Note that although the amount of information varies from line to line, there is no significant difference in the amount of information between the anomaly line and other lines, so simply comparing the amount of information does not provide an accurate description of the anomaly line. In a finite-order Markov model, the current state is solely related to the preceding state and is not influenced by earlier states. This is analogous to the grayscale correlation between pixels in hyperspectral images, being affected only by a specific spatial range. To make full use of the inter-line information relationship, this article designs an anomaly line determination method based on the finite Markov model [35]. Specifically,

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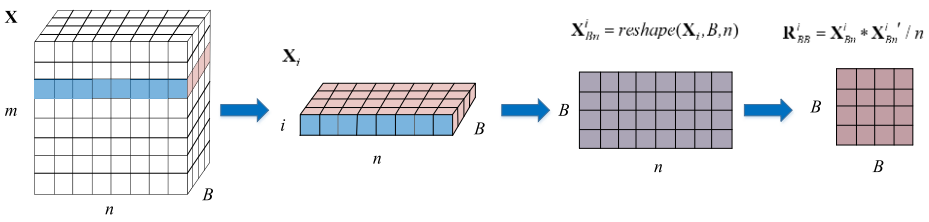


Fig. 2. Schematic of the data conversion principle of the line information of hyperspectral images.

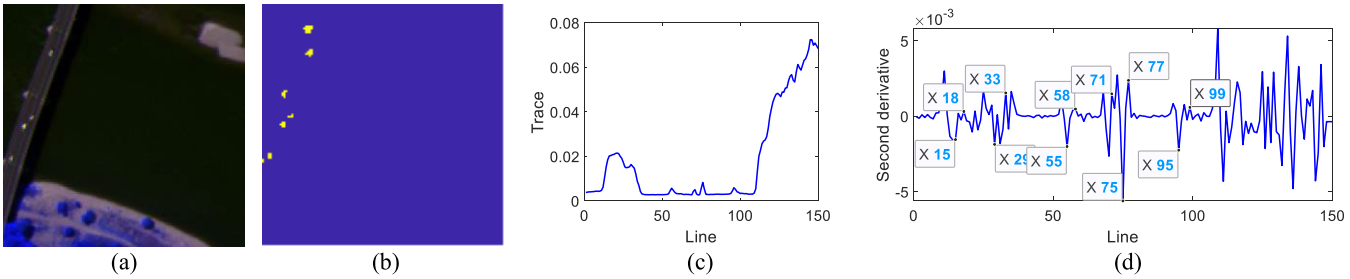


Fig. 3. Pavia Centre dataset. (a) Pseudocolor images. (b) Ground-truth map. (c) Line information. (d) Second derivative of the line information.

when the push-broom hyperspectral imaging sensor continuously obtains lines, the information of the current line is calculated, and the second derivative of the information of the obtained line is performed. The maximum and minimum of the second derivative values of the current line and the previous lines can be obtained by comparing the second derivative values of the information of the lines. Where, the minimal point of the second derivative of the information volume is the convexity of the information volume function reaches its maximal value, and the maximal point of the second derivative is the concavity of the information volume function reaches its maximum value, that is, the minimum and the maximal point corresponds to the inflection point of the function. This article defines the line connecting the two inflection points as the range where the anomaly line is located and takes the maximum point of the second derivative as the definite endpoint of the finite Markov model. It should be noted that the design of the inflection point identification method may misjudge some lines with rich background information as anomaly lines. In order to determine the change of inter-line information more accurately, the mean value and standard deviation on the basis of the second derivative of information are used. In the judgment process, if the information difference between the current line and the previous line exceeds the mean value, it reflects that the current line's information and the data's changing trend are significantly different. At the same time, if the information standard deviation from the current line to the initial line is higher than the information standard deviation from the previous line to the initial line, it indicates that the current line has a large variation within the dataset, thus increasing the order of the finite order. Taking the Pavia Centre dataset as an example, Fig. 3(d) is the second derivative value diagram of the information volume, where the marked data points are marked on the line where the anomaly line is located based on the finite Markov model, that is, the inflection point. Combined with the ground truth of the Pavia Centre dataset in Fig. 3(b), it can be seen that the marked points are the lines where the anomalies are located, indicating that the designed method is accurate.

C. Finite Markov Local Real-Time Correlation Matrix R Anomaly Detection (FMLRT-RAD)

Line-by-line real-time anomaly detection requires the push-broom hyperspectral imaging sensor to output the anomaly response values of all pixels in the line immediately after acquiring new scanning line data. The data processing process is shown in Fig. 4. When a large data sample is obtained, the data sample obtained at the beginning has little relation to the newly obtained line data, so using the entire data sample for background suppression is not appropriate for the current line. In addition, there are accumulated errors in the inverse identity of the existing line-by-line real-time anomaly detection calculation, which will also affect the detection results. To avoid the influence of redundant data and the stability of detection results, a new FMLRT-RAD is designed in this article. It is worth noting that different from the traditional line-by-line real-time anomaly detection, which uses all the acquired data for background suppression, the method proposed in this article is based on the finite Markov idea, determines the scope of the anomaly lines to use part of the obtained data samples, and completes the recursive update of relevant statistical information in the image. The data sample usage process is shown in Fig. 4. The line composed of dark green pixels is the line of pixels obtained at the current moment (the scanned line to be tested), and the white pixels are the line data not yet obtained. With data acquisition, light green pixels are used to represent the data that has been collected and processed by anomaly detection. According to the proposed FMLRT-RAD algorithm, gray pixels are the data samples that have been collected but need to be deleted, as shown in Fig. 4. During the process of push-broom hyperspectral

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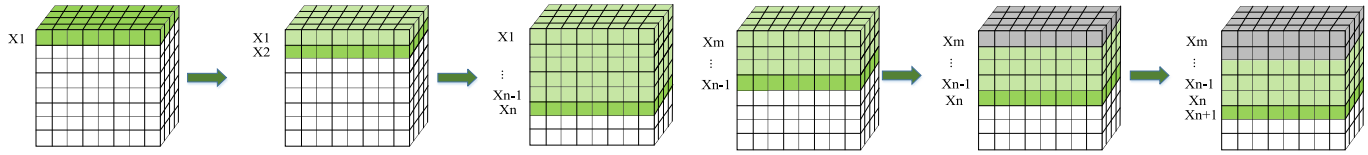


Fig. 4. Data sample model based on line-by-line processing.

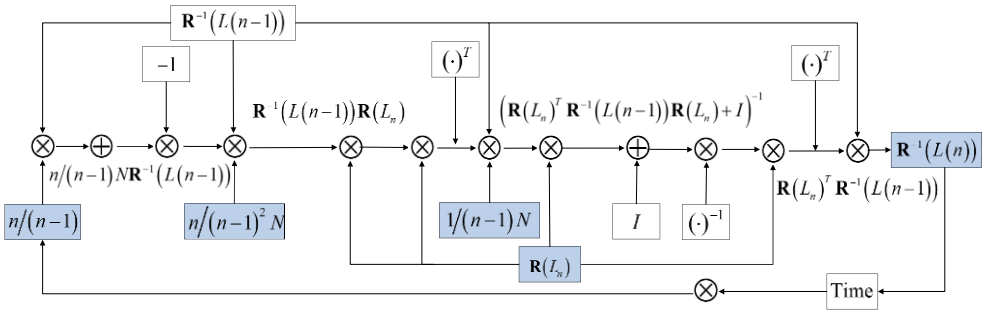


Fig. 5. Schematic of $\mathbf{R}^{-1}(L(n))$ in (12).

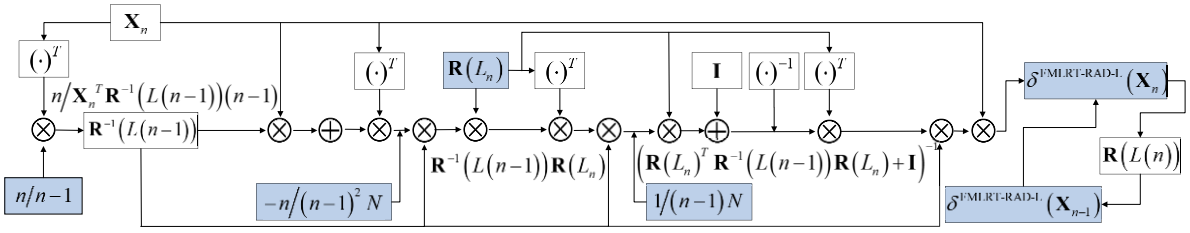


Fig. 6. Schematic of $\delta_n^{\text{FMLRT-RAD-L}}(\mathbf{X}_n)$ in (13).

imaging sensors, the data will be obtained line-by-line, and processed in real-time to output the all pixels in each scanning line. The detection operator according to R-AD is shown in the following:

$$\delta^{\text{R-AD}}(\mathbf{r}) = (\mathbf{r})^T(\mathbf{R})^{-1}\mathbf{r} \tag{10}$$

where $\mathbf{r}$ is the sample vector and $\mathbf{R}$ is the correlation matrix. According to different line information conditions, different calculation methods should be used. This article designs two different representation methods for correlation matrices based on the finite Markov model. Specifically, FMLRT-RAD is divided into two ways: anomaly line and nonanomaly line. The following will provide a detailed introduction.

1) FMLRT-RAD Using All Line Data (FMLRT-RAD-L): Using a finite Markov model with inter-line information in hyperspectral images, when no anomalies are found in the nth line of the current hyperspectral image, the correlation matrix of the data samples in the nth line of the hyperspectral image can be labeled as $\mathbf{R}(L(n))$, and its expression is shown in the following:

$$\begin{aligned}\mathbf{R}(L(n)) &= \frac{1}{nN}\left[\sum_{i=1}^{n-1}\mathbf{X}_i\mathbf{X}_i^T + \mathbf{X}_n\mathbf{X}_n^T\right] \\ &= \left(\frac{n-1}{n}\right)\mathbf{R}(L(n-1)) + \frac{1}{nN}\mathbf{R}(L_n)(\mathbf{R}(L_n))^T.\end{aligned} \tag{11}$$

Among them, $L(n)$ represents the first n lines of data samples, $\mathbf{X}_i$ represents the ith line of data samples, $\mathbf{X}_n$ represents the nth line of data samples, and $\mathbf{R}(L_n)$ represents the correlation matrix of the nth line of data samples in the hyperspectral image.

To solve (11), according to the Woodbury identity $(\mathbf{A}+\mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{DA}^{-1}\mathbf{B}+\mathbf{C}^{-1})^{-1}\mathbf{DA}^{-1}$, let $\mathbf{A} = (n-1/n)\mathbf{R}(L(n-1))$, $B = (\mathbf{R}(L_n)/(nN)^{1/2})$, $\mathbf{C} = \mathbf{I}$, $D = ((\mathbf{R}(L_n)/\sqrt{nN}))^T$, then the result of $\mathbf{R}(L(n))$ inversion is expressed as (12), as shown at the bottom of the next page.

By combining (12) with (11), FMLRT-RAD-L can be derived to achieve real-time anomaly detection of the first line of data samples.

Among them, $\mathbf{X}_n = \left[\mathbf{x}_1^n, \mathbf{x}_2^n, \ldots, \mathbf{x}_N^n\right]$ is the nth line of data in the hyperspectral image, $\mathbf{x}_i^n$ is the ith pixel, and N is all the pixels in each line. Fig. 5 is a schematic of implementing $\mathbf{R}^{-1}(L(n))$ (12), where Time is a time delay. It is worth noting that when no anomalies are found in the nth line of the hyperspectral image, FMLRT-RAD-L also uses the inverse $\mathbf{R}^{-1}(L(n))$ (12) of the derived correlation matrix to achieve spectral statistics $\delta_n^{\text{FMLRT-RAD-L}}(\mathbf{X}_n)$ (13), as shown at the bottom of the next page of the current data sample $\mathbf{X}_n$, as shown in Fig. 6.

2) FMLRT-RAD Using One-In and One-Out Line Data (FMLRT-RAD-IOL): Using the finite Markov model of inter-line information of hyperspectral images, when an anomaly is

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found in the nth line of the current hyperspectral image, the correlation matrix of the data samples in the first $n-1$ lines of the hyperspectral image can be marked as $\mathbf{R}(L(\text{new}))$, and its expression is shown in the following:

$$\begin{aligned}&\mathbf{R}(L(\text{new}))\\&=\frac{1}{nN}\sum_{i=m+1}^{n}\mathbf{X}_i\mathbf{X}_i^T=\frac{1}{nN}\left[\sum_{i=m+1}^{n-1}\mathbf{X}_i\mathbf{X}_i^T+\mathbf{X}_n\mathbf{X}_n^T-\mathbf{X}_m\mathbf{X}_m^T\right]\\&=\mathbf{R}(L(n-1))+\frac{1}{n}\mathbf{R}(L_n)(\mathbf{R}(L_n))^T-\frac{1}{n}\mathbf{R}(L_m)(\mathbf{R}(L_m))^T.\end{aligned}\tag{14}$$

Using $(\mathbf{A}+\mathbf{BCD})^{-1}=\mathbf{A}^{-1}-\mathbf{A}^{-1}\mathbf{B}(\mathbf{DA}^{-1}\mathbf{B}+\mathbf{C}^{-1})^{-1}\mathbf{DA}^{-1}$, first divide $\mathbf{R}(L(\text{new}))$ into two parts, $\mathbf{R}(L(\text{new}))=\mathbf{R}^*-(1/n)\mathbf{R}(L_m)(\mathbf{R}(L_m))^T$ and $\mathbf{R}^*=\mathbf{R}(L(n-1))+(1/n)\mathbf{R}(L_n)(\mathbf{R}(L_n))^T$, compute $\mathbf{R}^*$ first, then calculate $\mathbf{R}(L(\text{new}))$, and finally plug $\mathbf{R}^*$ into $\mathbf{R}(L(\text{new}))$.

Calculate $\mathbf{R}^{*-1}$ first, let $\mathbf{A}=\mathbf{R}(L(n-1))$, $B=(\mathbf{R}(L_n)/\sqrt{n})$, $\mathbf{C}=\mathbf{I}$, $D=((\mathbf{R}(L_n)/\sqrt{n}))^T$, according to $(\mathbf{A}+\mathbf{BCD})^{-1}=\mathbf{A}^{-1}-\mathbf{A}^{-1}\mathbf{B}(\mathbf{DA}^{-1}\mathbf{B}+\mathbf{C}^{-1})^{-1}\mathbf{DA}^{-1}$, (15), as shown at the bottom of the next page, can be obtained. And then calculate $\mathbf{R}^{-1}(L(\text{new}))$, let $\mathbf{A}=\mathbf{R}^*$, $B=(\mathbf{R}(L_m)/\sqrt{n})$, $\mathbf{C}=\mathbf{I}$, $D=((\mathbf{R}(L_m)/\sqrt{n}))^T$, according to $(\mathbf{A}-\mathbf{BCD})^{-1}=\mathbf{A}^{-1}+\mathbf{A}^{-1}\mathbf{B}(\mathbf{DA}^{-1}\mathbf{B}-\mathbf{C}^{-1})^{-1}\mathbf{DA}^{-1}$, (16), as shown at the bottom of the next page, can be obtained. Then combine (15) with (16) to get (17), as shown at the bottom of the next page. Combining (17) with (11) can derive FMLRT-RAD-IOL to achieve real-time anomaly detection of the nth line of data samples.

Fig. 7 is a schematic of implementing $\mathbf{R}^{*-1}$ (15), Fig. 8 is a schematic of separately implementing $\mathbf{R}^{-1}(L(\text{new}))$ (16), and Fig. 9 is a schematic of combining (15) with (16) to achieve (17), where Time is a time delay. It is worth noting that when an anomaly is found in the nth line of the current hyperspectral image, FMLRT-RAD-IOL also uses the inverse (17) of the derived correlation matrix to achieve $\delta_n^{\text{FMLRT-RAD-IOL}}(\mathbf{X}_n)$ (18), as shown at the bottom of the next page, for the spectral statistics of the current data sample $\mathbf{X}_n$, as shown in Fig. 10.

A detailed step-by-step implementation of FMLRT-RAD is shown in Algorithm 1.

III. Computational Complexity and Time Complexity

The computational complexity and time complexity are important indicators for evaluating the performance of real-time local anomaly detection algorithms, which can reflect the efficiency, real-time performance, and applicability of the algorithm. Table I provides detailed information on the computational complexity of the proposed FMLRT-RAD method,

$$\begin{aligned}&\mathbf{R}^{-1}(L(n))\\&=\left(\left(\frac{n-1}{n}\right)\mathbf{R}(L(n-1))+\frac{\mathbf{R}(L_n)}{\sqrt{nN}}\mathbf{I}\left(\frac{\mathbf{R}(L_n)}{\sqrt{nN}}\right)^T\right)^{-1}\\&=\left(\frac{n}{n-1}\right)\mathbf{R}^{-1}(L(n-1))-\left(\frac{n}{n-1}\right)\mathbf{R}^{-1}(L(n-1))\frac{\mathbf{R}(L_n)}{\sqrt{nN}}\left(\left(\frac{\mathbf{R}(L_n)}{\sqrt{nN}}\right)^T\left(\frac{n}{n-1}\right)\mathbf{R}^{-1}(L(n-1))\frac{\mathbf{R}(L_n)}{\sqrt{nN}}+\mathbf{I}\right)^{-1}\\&\quad\times\left(\frac{\mathbf{R}(L_n)}{\sqrt{nN}}\right)^T\left(\frac{n}{n-1}\right)\mathbf{R}^{-1}(L(n-1))\\&=\left(\frac{n}{n-1}\right)\mathbf{R}^{-1}(L(n-1))-\left(\frac{n}{(n-1)^2N}\right)\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\left(\frac{1}{(n-1)N}\right)(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}*\\&(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\end{aligned}\tag{12}$$

$$\begin{aligned}&\delta_n^{\text{FMLRT-RAD-L}}(\mathbf{X}_n)\\&=\mathbf{X}_n^T\mathbf{R}^{-1}(L(n))\mathbf{X}_n\\&=\mathbf{X}_n^T\left\{\begin{array}{l}\left(\frac{n}{n-1}\right)\mathbf{R}^{-1}(L(n-1))-\left(\frac{n}{(n-1)^2N}\right)\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\left(\frac{1}{(n-1)N}\right)(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}*(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\end{array}\right\}\mathbf{X}_n\\&=\left(\frac{n}{n-1}\right)\mathbf{X}_n^T\mathbf{R}^{-1}(L(n-1))\mathbf{X}_n\\&\quad-\mathbf{X}_n^T\left\{\begin{array}{l}\left(\frac{n}{(n-1)^2N}\right)\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\left(\frac{1}{(n-1)N}\right)(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}*(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\end{array}\right\}\mathbf{X}_n\\&=\delta_{n-1}^{\text{FMLRT-RAD-L}}(\mathbf{X}_{n-1})-\frac{n}{(n-1)^2N}\mathbf{X}_n^T\left\{\begin{array}{l}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\left(\frac{1}{(n-1)N}\right)(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}*(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\end{array}\right\}\mathbf{X}_n\end{aligned}\tag{13}$$

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where the third column represents the mathematical complexity of the second column in textual form, and the fourth column represents this complexity. Table I helps to better understand the execution of the proposed algorithm on the hyperspectral image data.

Matrix inversion usually involves computationally complex operations. Table II compares the computational complexity of the R-AD and the proposed FMLRT-RAD method for processing a single signature. It can be seen that different from the R-AD method complexity $O(n^3)$, the complexity of the proposed method is only $O(n)$. This is because the FMLRT-RAD method adopts the recursive update formula of Woodbury identity, and uses the low-complexity addition and subtraction operations of the matrix in the processing to

$$
\begin{aligned}
&\mathbf{R}^{*-1}\\
&=\left(\mathbf{R}(L(n-1))+\frac{\mathbf{R}(L_n)}{\sqrt{n}}\mathbf{I}\left(\frac{\mathbf{R}(L_n)}{\sqrt{n}}\right)^T\right)^{-1}\\
&=\mathbf{R}^{-1}(L(n-1))-\mathbf{R}^{-1}(L(n-1))\frac{\mathbf{R}(L_n)}{\sqrt{n}}\left(\left(\frac{\mathbf{R}(L_n)}{\sqrt{n}}\right)^T\mathbf{R}^{-1}(L(n-1))\frac{\mathbf{R}(L_n)}{\sqrt{n}}+\mathbf{I}\right)^{-1}\left(\frac{\mathbf{R}(L_n)}{\sqrt{n}}\right)^T\mathbf{R}^{-1}(L(n-1))\\
&=\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))
\end{aligned}
\tag{15}
$$

$$
\begin{aligned}
&\mathbf{R}^{-1}(L(\text{new}))\\
&=\left(\mathbf{R}^*-\frac{1}{n}\mathbf{R}(L_m)(\mathbf{R}(L_m))^T\right)^{-1}=\mathbf{R}^{*^{-1}}+\mathbf{R}^{*^{-1}}\frac{\mathbf{R}(L_m)}{\sqrt{n}}\left(\left(\frac{\mathbf{R}(L_m)}{\sqrt{n}}\right)^T\mathbf{R}^{*^{-1}}\frac{\mathbf{R}(L_m)}{\sqrt{n}}-\mathbf{I}\right)^{-1}\left(\frac{\mathbf{R}(L_m)}{\sqrt{n}}\right)^T\mathbf{R}^{*^{-1}}\\
&=\mathbf{R}^{*^{-1}}+\frac{1}{n}\mathbf{R}^{*^{-1}}\mathbf{R}(L_m)\left(\frac{1}{n}(\mathbf{R}(L_m))^T\mathbf{R}^{*^{-1}}\mathbf{R}(L_m)-\mathbf{I}\right)^{-1}(\mathbf{R}(L_m))^T\mathbf{R}^{*^{-1}}
\end{aligned}
\tag{16}
$$

$$
\begin{aligned}
&\mathbf{R}^{-1}(L(\text{new}))\\
&=\mathbf{R}^{*^{-1}}+\frac{1}{n}\mathbf{R}^{*^{-1}}\mathbf{R}(L_m)\left(\frac{1}{n}(\mathbf{R}(L_m))^T\mathbf{R}^{*^{-1}}\mathbf{R}(L_m)-\mathbf{I}\right)^{-1}(\mathbf{R}(L_m))^T\mathbf{R}^{*^{-1}}\\
&=\left(\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\right)\\
&\quad+\frac{1}{n}\left(\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\right)\\
&\quad*\mathbf{R}(L_m)\left(\frac{1}{n}(\mathbf{R}(L_m))^T\left(\begin{array}{l}\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)*\\ \left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\end{array}\right)\mathbf{R}(L_m)-\mathbf{I}\right)^{-1}*(\mathbf{R}(L_m))^T\\
&\quad*\left(\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\right)
\end{aligned}
\tag{17}
$$

$$
\begin{aligned}
&\delta_n^{\text{FMLRT-RAD-IOL}}(\mathbf{X}_n)\\
&=\mathbf{X}_n^T\mathbf{R}^{-1}(L(\text{new}))\mathbf{X}_n\\
&=\delta_{n-1}^{\text{FMLRT-RAD-IOL}}(\mathbf{X}_{n-1})-\frac{1}{n}\mathbf{X}_n^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{X}_n\\
&\quad+\frac{1}{n}\mathbf{X}_n^T\left(\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\right)\\
&\quad*\mathbf{R}(L_m)\left(\frac{1}{n}(\mathbf{R}(L_m))^T\left(\begin{array}{l}\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\\ *\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\end{array}\right)\mathbf{R}(L_m)-\mathbf{I}\right)^{-1}*(\mathbf{R}(L_m))^T\\
&\quad*\left(\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\right)\mathbf{X}_n
\end{aligned}
\tag{18}
$$

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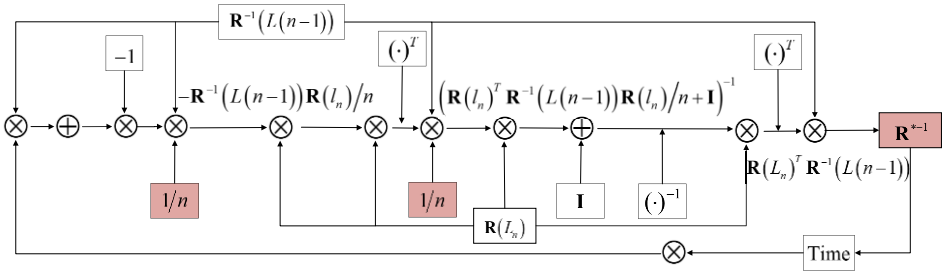


Fig. 7. Schematic of $\mathbf{R}^{*-1}$ in (15).

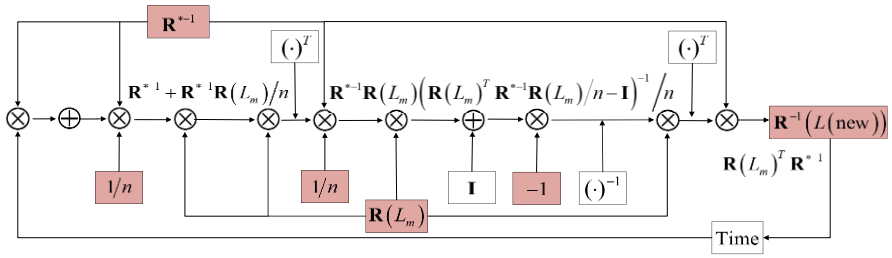


Fig. 8. Schematic of $\mathbf{R}^{-1}(L(new))$ in (16).

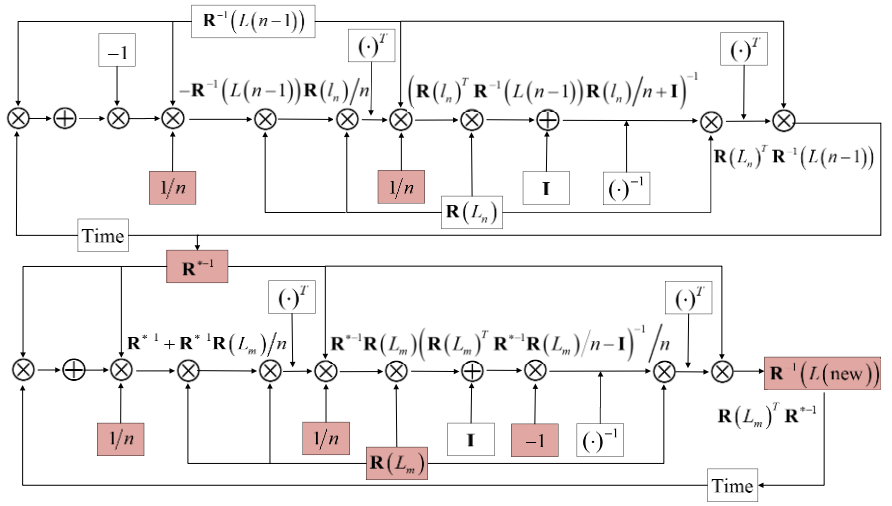


Fig. 9. Schematic of $\mathbf{R}^{-1}(L(new))$ in (17).

avoid the calculation of matrix inversion, thus improving the computational efficiency.

To verify the effectiveness of the proposed FMLRT-RAD algorithm in terms of real-time performance, an experiment on real hyperspectral image data is conducted and compared with the R-AD algorithm [11] and the RT-CR-RXD algorithm [25]. The experimental results are shown in Fig. 11. The R-AD algorithm is chosen for comparison because it is a typical anomaly detection algorithm, which is helpful to evaluate the performance of the FMLRT-RAD algorithm on anomaly detection tasks. The RT-CR-RXD algorithm is a real-time anomaly detection method designed for line-by-line data acquisition. By comparing these two algorithms with the FMLRT-RAD algorithm, we can better evaluate the real-time performance capability. In Fig. 3, there is some information about the Pavia Centre dataset, and in the experimental design, we detailed the details of it. Fig. 11 shows the single-line calculation time required for the three methods as the number of different data lines increases from 1 to 150. In Fig. 11, the X-axis represents the number of data lines, while the Y-axis represents the calculation time.

According to the result in Table III, it can be seen that the FMLRT-RAD method is superior to the R-AD algorithm, and comparable to the real-time RT-CR-RXD algorithm in calculation time. It is noteworthy that the proposed FMLRT-RAD method, while slightly more time-consuming than the RT-CR-RXD algorithm, outweighs it in terms of accuracy, as detailed in Figs. 15–19 and Tables IV–VIII in Section V. The RT-CR-RXD algorithm requires the utilization of all preceding data, encompassing substantial interference and irrelevant background elements, introducing cumulative errors through the inversion of the identity equation, thus impacting the accuracy of detection outcomes. Consequently, given the paramount

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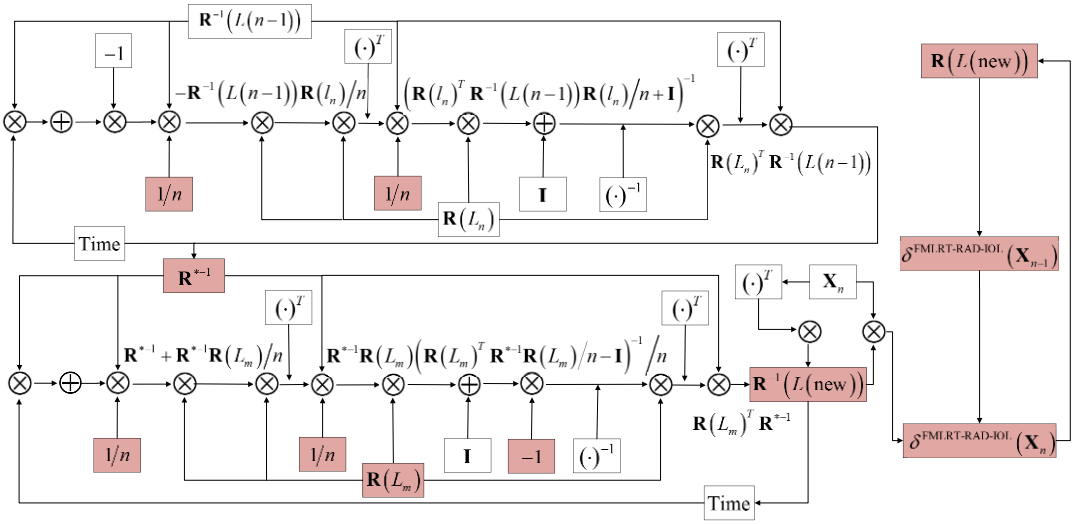


Fig. 10. Schematic of $\delta_n^{\text{FMLRT-RAD-IOL}}(\mathbf{X}_n)$ in (18).

TABLE I
COMPUTATIONAL COMPLEXITY OF (18) TO CALCULATE

Innovations information: $\mathbf{R}(\text{L(new)})$ by (14)	$\mathbf{R}(\text{L(new)})=\mathbf{R}(L(n-1))+\frac{1}{n}\mathbf{R}(L_n)(\mathbf{R}(L_n))^T-\frac{1}{n}\mathbf{R}(L_m)(\mathbf{R}(L_m))^T$	One inner product and inner products of two n-dimensional vectors plus 1 subtraction and 1 multiplication	$O(n)$
Innovations information: $\mathbf{R}^{*\text{-}1}$ by (15)	$\mathbf{R}^{*\text{-}1}=\mathbf{R}^{-1}(L(n-1))-$ $\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))$	One inner product and seven inner products plus 1 subtraction and 1 multiplication	$O(n)$
Innovations information: $\mathbf{R}^{-1}(\text{L(new)})$ by (17)	$\mathbf{R}^{-1}(\text{L(new)})=\mathbf{R}^{*-1}+\frac{1}{n}\mathbf{R}^{*-1}\mathbf{R}(L_m)\left(\frac{1}{n}(\mathbf{R}(L_m))^T\mathbf{R}^{*-1}\mathbf{R}(L_m)-\mathbf{I}\right)^{-1}(\mathbf{R}(L_m))^T\mathbf{R}^{*-1}$	One inner product and seven inner products plus 1 multiplication and 1 subtraction	$O(n)$
Update $\delta_n^{\text{FMLRT-RAD-IOL}}(\mathbf{X}_n)$ by (14), (15), and (17).	$\delta_n^{\text{FMLRT-RAD-IOL}}(\mathbf{X}_n)$ $=\delta_{n-1}^{\text{FMLRT-RAD-IOL}}(\mathbf{X}_{n-1})-\frac{1}{n}\mathbf{X}_n^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{X}_n+$ $\frac{1}{n}\mathbf{X}_n^T\left(\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\right)*$ $\mathbf{R}(L_m)\left(\frac{1}{n}(\mathbf{R}(L_m))^T\mathbf{R}^{*-1}\mathbf{R}(L_m)-\mathbf{I}\right)^{-1}*(\mathbf{R}(L_m))^T*$ $\left(\mathbf{R}^{-1}(L(n-1))-\frac{1}{n}\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)\left(\frac{1}{n}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\mathbf{R}(L_n)+\mathbf{I}\right)^{-1}(\mathbf{R}(L_n))^T\mathbf{R}^{-1}(L(n-1))\right)\mathbf{X}_n$	Inner products of two n-dimensional vectors plus nine inner products plus 1 subtraction and 1 multiplication plus twenty-one inner products and 1 subtraction and 1 multiplication	$O(n)$

TABLE II
COMPARISON OF COMPUTATION COMPLEXITY OF MATRIX INVERSIONS BY PROCESSING A SINGLE SIGNATURE

	R-AD specified by (10)	FMLRT-RAD specified by (18)
Input	$\mathbf{U}_1,\mathbf{U}_2,...,\mathbf{U}_n$	$\mathbf{U}$
The complexity of the method	$O(n^3)$	$O(n)$

importance of detection accuracy, the computational investment within finite-order time frames for the proposed method is justified. Overall, the calculation time required by the FMLRT-RAD method tends to stabilize as the number of lines increases, and the processing time per signal source remains almost constant. The experimental results clearly show the excellent real-time performance of the FMLRT-RAD algorithm in processing real hyperspectral image data.

IV. EXPERIMENT DESIGN

A. Real Hyperspectral Image Anomaly Detection Datasets

The performance of the FMLRT-RAD algorithm is evaluated on five different image datasets, which are.

1) *Segundo Dataset:* The first dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [36]. An area consisting of an oil refinery, several residential areas, parks, and a school district in El Segundo, CA, USA, is described, which includes 224 spectral bands in the 366–2469 nm range. The image space is 250 × 300, contains 2048 anomaly

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Algorithm 1 FMLRT-RAD

Input: Hyperspectral image.

Initialize: Calculate the information of the current hyperspectral image line H_i, the second derivative value of the information $ddy(H_i)$, the mean of the information from the current line to the initial line Mean(H_i), and the standard deviation of the information from the current line to the initial line STD(H_i).

If $ddy_{i-1} > 0$, $ddy_i < 0$, $H_i - H_{i-1} >$ Mean(H_{i-1}), STD$(H_i) >$ STD(H_{i-1}), calculate R by (14).

Update the inverse of the correlation matrix R, calculate by (17).

Compute the anomalous degree of each pixel: calculate $\delta^{\text{FMLRT-RAD}}(\mathbf{r}_n^m)$ by (18).

Elseif $ddy_{i-1} < 0, ddy_i > 0$, $H_i - H_{i-1} >$ Mean(H_{i-1}), STD$(H_i) >$ STD(H_{i-1}), calculate R by (14).

Update the inverse of the correlation matrix R, calculate by (17).

Compute the anomalous degree of each pixel: calculate $\delta^{\text{FMLRT-RAD}}(\mathbf{r}_n^m)$ by (18).

Else calculate R by (11).

Update the inverse of the correlation matrix R, calculate R^{-1} by (12).

Compute the anomalous degree of each pixel: calculate $\delta^{\text{FMLRT-RAD}}(\mathbf{r}_n^m)$ by (13).

End

Update n: $n \leftarrow n+1$

Output: Anomaly map.

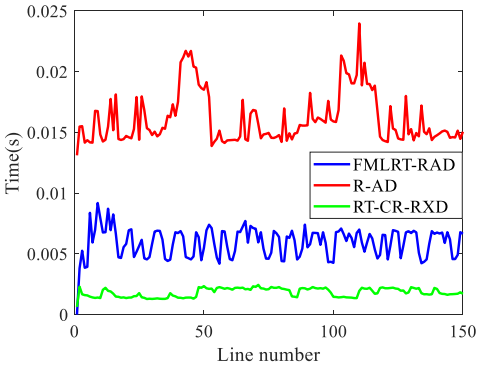


Fig. 11. Comparison of calculation time required by FMLRT-RAD and other algorithms in Pavia Centre dataset.

pixels, and has a ground resolution of 7.1 m. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 12(a).

2) *Pavia Centre Dataset:* The second dataset was obtained using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in Pavia, Italy [37]. The image mainly includes bridges, water, exposed soil, and buildings, with vehicles on the bridges seen as anomalies. The whole image contains 1096 × 1096 pixels, and a scene of 150 × 150 pixels is selected for anomaly detection, including 68 anomaly pixels, with a spatial resolution of 1.3 m. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 12(b).

TABLE III

COMPARATIVE PERFORMANCE OF AVERAGE COMPUTING TIMES REQUIRED BY FMLRT-RAD, R-AD, AND RT-CR-RXD

Method	Pavia Centre Dataset
FMLRT-RAD (seconds)	**0.0059**
R-AD (seconds)	0.0161
RT-CR-RXD (seconds)	**0.0018**

TABLE IV

AUC VALUES OF DIFFERENT METHODS ON SEGUNDO

Method	AUC of (P_D, P_F)	AUC of (P_D, τ)	AUC of (P_F, τ)
FMLRT-RAD	**0.9364**	**0.0483**	0.0105
RXD	0.5767	0.0113	**0.0049**
R-AD	0.5741	0.0112	**0.0049**
RT-CR-RXD	**0.9229**	**0.0303**	0.0061
PBSFRT	0.5741	0.0112	**0.0049**
LSAD-CR-IDW	0.5236	0.0082	**0.0049**
FEBPAD	0.6739	0.0162	0.0053
VABS	0.6756	0.0293	0.0051
RGAE	0.8269	0.0249	**0.0049**
CRDBPSW	0.6646	0.0139	0.0051
AHMID	0.5663	0.0152	**0.0050**

TABLE V

AUC VALUES OF DIFFERENT METHODS ON PAVIA CENTRE

Method	AUC of (P_D, P_F)	AUC of (P_D, τ)	AUC of (P_F, τ)
FMLRT-RAD	**0.9784**	**0.3193**	0.0829
RXD	0.9485	0.1289	0.0231
R-AD	0.9438	0.1279	0.0273
RT-CR-RXD	**0.9622**	0.1617	0.0342
PBSFRT	0.9438	0.1279	0.0273
LSAD-CR-IDW	0.9370	**0.3885**	0.1630
FEBPAD	0.9446	0.1601	0.0174
VABS	0.9313	0.0961	**0.0055**
RGAE	0.9015	0.2183	0.0409
CRDBPSW	0.8858	0.1786	0.0623
AHMID	0.9640	0.1412	**0.0121**

TABLE VI

AUC VALUES OF DIFFERENT METHODS ON ABU

Method	AUC of (P_D, P_F)	AUC of (P_D, τ)	AUC of (P_F, τ)
FMLRT-RAD	**0.9164**	**0.2657**	0.0866
RXD	0.8337	0.1826	0.0511
R-AD	0.8291	0.1823	0.0516
RT-CR-RXD	**0.9110**	0.2122	0.0592
PBSFRT	0.8291	0.1823	0.0516
LSAD-CR-IDW	0.9013	**0.3063**	0.1526
FEBPAD	0.5387	0.0106	0.0183
VABS	0.8841	0.2318	**0.0093**
RGAE	0.7119	0.0757	**0.0200**
CRDBPSW	0.8750	0.1939	0.0418
AHMID	0.9069	0.1699	0.0361

3) *Airport-Beach-Urban (ABU) Dataset:* The third ABU dataset was taken by AVIRIS sensor in Los Angeles and two aircraft were considered anomalies [38]. Contains 205 bands, the image space size is 100 × 100, contains 87 anomaly pixels, and the spatial resolution is 7.1 m. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 12(c).

4) *Los Angeles Dataset:* The fourth dataset was the Los Angeles scene collected by the AVIRIS sensor [39]. It has 205 bands in the 430–860 wavelength range and a spatial resolution of 7.1 m. The space size of the entire

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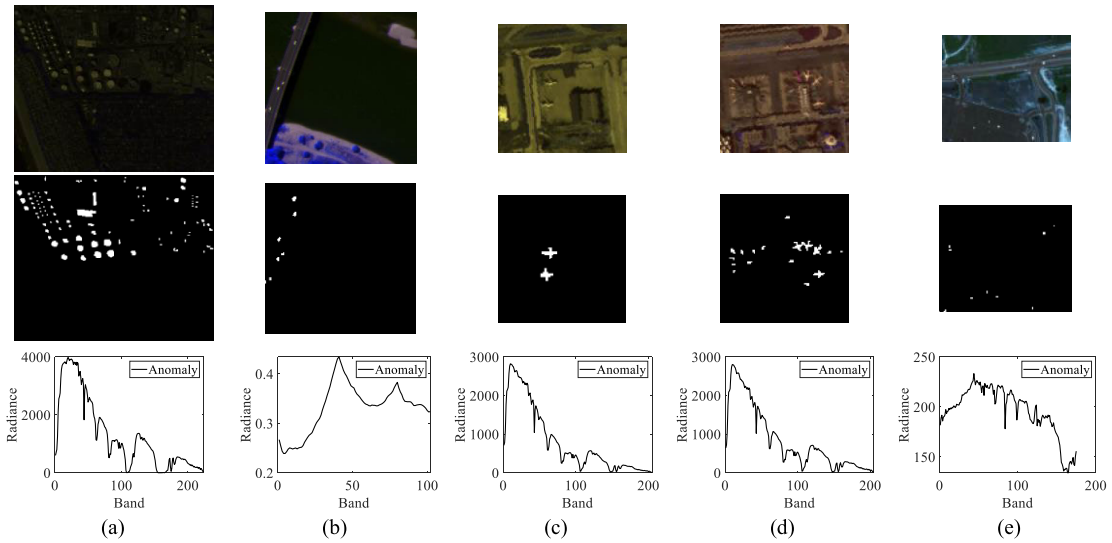


Fig. 12. Pseudocolor images, ground-truth maps, and spectral signature of five experimental datasets. (a) Segundo. (b) Pavia Centre. (c) ABU. (d) Los Angeles. (e) HYDICE Urban.

TABLE VII
AUC VALUES OF DIFFERENT METHODS IN LOS ANGELES

Method	AUC of (P_D, P_F)	AUC of (P_D, τ)	AUC of (P_F, τ)
FMLRT-RAD	**0.9279**	**0.1959**	0.0559
RXD	0.9021	0.0643	0.0141
R-AD	0.9005	0.0637	0.0140
RT-CR-RXD	**0.9249**	**0.1489**	0.0346
PBSFRT	0.9005	0.0637	0.0140
LSAD-CR-IDW	0.9100	0.0864	0.0369
FEBPAD	0.8951	0.0931	**0.0165**
VABS	0.7548	0.0370	**0.0058**
RGAE	0.8220	0.0493	0.0077
CRDBPSW	0.6568	0.0362	0.0057
AHMID	0.4944	0.0109	**0.0049**

TABLE VIII
AUC VALUES OF DIFFERENT METHODS ON HYDICE URBAN

Method	AUC of (P_D, P_F)	AUC of (P_D, τ)	AUC of (P_F, τ)
FMLRT-RAD	**0.9854**	**0.3860**	0.0672
RXD	0.9818	0.2331	0.0350
R-AD	0.9819	0.2298	0.0348
RT-CR-RXD	**0.9849**	0.3179	0.0494
PBSFRT	0.9819	0.2298	0.0348
LSAD-CR-IDW	0.9529	**0.4671**	0.0797
FEBPAD	0.9171	0.1388	0.0201
VABS	0.7836	0.0898	**0.0131**
RGAE	0.7575	0.1917	0.0988
CRDBPSW	0.8621	0.2757	0.0296
AHMID	0.9745	0.3271	**0.0140**

scene is 100 × 100, including 170 anomaly pixels, and the aircraft in the scene is an anomaly. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 12(d).

5) *HYDICE Urban Dataset:* The fifth dataset was published by Hyperspectral Digital Imagery Collection Experiment (HYDICE) [40]. The scene is mainly composed of highways, factories, roofs, and vehicles, and the vehicle target is the main anomaly. The original scene contains 307 × 307 pixels, and the subregion with the size of 80 × 100 pixels is selected for anomaly detection, including 21 anomaly pixels, 175 bands in the spectral range of 440–2370 nm, and the spatial resolution is 1 m. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 12(e).

The focus of this study is the real-time anomaly detection of hyperspectral images, so the real-world datasets that are widely used in anomaly detection are used in the experiments [41], [42], [43], [44]. To reflect the detection effect of the proposed method on different data, the scene sizes and anomalies of the five datasets are different.

B. Comparison Methods and Parameter Settings

The comparison methods include the proposed classical RX-based methods, real-time anomaly detection methods, and state-of-the-art hyperspectral image anomaly detection methods in the recent three years on these datasets.

Two classical RX-based methods are used for comparisons on all datasets to show the proposed RX-based validity. Among them, RXD [10] is the benchmark of anomaly detection methods. R-AD [11] is an anomaly detection method based on the correlation matrix R. By comparing these two methods, the response degree of the proposed algorithm to the anomaly target can be seen.

To verify the real-time performance, the proposed method is compared with RT-CR-RXD [25] for line-by-line processing, and progressive band subset fusion real time (PBSFRT) [44] for band-by-band processing. It is worth noting that the method of obtaining data in this article is line-by-line, and it is an anomaly detection method based on correlation matrix R, which are same as RT-CR-RXD, so the real-time detection ability can be seen by comparing it with the real-time method.

And it is compared with the existing advanced methods on different datasets. These advanced methods use different strategies to detect anomalies and achieve satisfactory results on datasets. For example, LSAD-CR-IDW [20] uses the surrounding pixels collected from the outside window and the inside window to represent the test pixels linearly. In this article, the three parameters selected by the LSAD-CR-IDW algorithm are all set as $\text{win}_{\text{out}} = 5$, $\text{win}_{\text{in}} = 3$ and

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$\lambda = 0.01$ in the comparison experiment. Collaborative representation detector with background purificationand saliency weight (CRDBPSW) [21] using the least squares technique to refine the local background, and the parameter in this article is set to $\text{win}_{\text{out}} = 25$, $\text{win}_{\text{in}} = 15$ and $\lambda =$ 1e-6. The advantages of detection accuracy can be seen by comparing the proposed method with the surrounding pixels collected from the window. FEBPAD [14] uses row-constrained LRaSMD to separate background noise and anomalies, and its comparison can reflect the practicability and stability of detection results. VABS [13] realizes the noise reduction processing of background and noise of significant feature map through curvature filtering and obtains the initial anomaly region map, which is similar to the initial anomaly line range obtained by calculation of the proposed method, and the anomaly detection ability can be seen by comparing with it. RGAE [15] can automatically extract the inherent features of images, and its comparison can reflect the ability of the proposed method in detecting anomalies that are significantly different from other features. AHMID [17] uses structurally incoherent constraints to separate the background from the anomaly, and by comparing with them, it can be seen that the expression of the anomaly and the inhibition ability of the background can be seen. In general, the proposed algorithm is compared with ten different anomaly detection methods. It is worth noting that when verifying the applicability of the present method on five datasets, we simulated the data according to the acquisition method of the push-broom hyperspectral imaging sensor. However, nine of the ten comparison methods do not directly process the line-by-line data obtained but need to be detected after the full data is obtained. Therefore, the proposed algorithm is superior in real-time application of push-broom hyperspectral imaging system.

V. Experimental Results and Discussions

In this section, the performance of the FMLRT-RAD method is deeply analyzed. Five real hyperspectral data from different scenarios are used to evaluate the effectiveness and superiority of our proposed FMLRT-RAD method and compared the detection accuracy of FMLRT-RAD with existing methods. Then the detection diagram of each test method for qualitative evaluation of anomaly detection performance is provided. 3-D receiver operating characteristic curve (3-D ROC) [45], [46], a 2-D ROC [47], [48], and an area under the ROC curve (AUC) are used to conduct a quantitative evaluation of detection performance. In the 3-D ROC curve, a specific threshold parameter τ is the third independent variable and generates (P_D, P_F, τ) function to evaluate the anomaly detection performance. The 2-D ROC curve includes the 2-D curve of (P_D, P_F), the 2-D curve of (P_D, τ), and the 2-D curve of (P_F, τ). The AUC value reflects the area enclosed by the ROC curve and the axes. The higher the AUC values in (P_D, P_F), and (P_D, τ), the better the detectability of the anomaly. In the case of AUC (P_F, τ), the smaller the AUC value, the better the suppression effect on the background (BKG).

All the comparison algorithms were implemented using MATLAB R2020a on a PC equipped with Intel Core i7 CPU@1.8 GHz and 16 GB RAM.

A. Data Results

The detection process of FMLRT-RAD on five real hyperspectral datasets, including detection graphs and grayscale graphs, is shown in Fig. 13. The detection results obtained by FMLRT-RAD, the classical RX-based methods, real-time anomaly detection methods, and the advanced hyperspectral image anomaly detection methods in recent three years are shown in Fig. 14.

From Fig. 13, it can be seen that the proposed FMLRT-RAD proposed in this article can accurately detect the position and shape of different size anomalies in different scenarios, such as aircraft and vehicles. It should be noted that in the initial stage of detection, the image is generally lighter in color, but once the algorithm detects an anomaly, the background is suddenly suppressed, and the brightness value of the anomaly pixel is significantly increased. This phenomenon can be attributed to several factors. First, FMLRT-RAD obtains the initial exception range by analyzing the information of the current and previous lines. Second, according to the detected anomaly information, only some lines and the data of the current line are selected in the calculation process to generate the detection result. This local information processing method can not only retain anomalies well, but also effectively suppress background interference, and realize the accurate detection of different-size anomalies in different scenes.

The comparison of interference, and realize the accurate detection of the proposed method and other anomaly detection methods on five datasets is shown in Fig. 14. It can be seen that compared with other methods, the proposed FMLRT-RAD has a better anomaly detection effect.

Specifically, in the Segundo dataset, the FMLRT-RAD algorithm detects almost all anomalies and can realize real-time processing of the algorithm. However, RXD, RAD, PBSFRT, and FEBPAD methods only detect a small part of the anomalies and lose most of the anomalies. This is because all the sampled data are used to calculate the covariance matrix and correlation matrix, which can better suppress the background information, but the local anomaly targets are also drowned in the global background. The VABS, RGAE, CRDBPSW, and AHMID methods also show a lack of detection of some anomalies in the middle part of the image. Interestingly, the advantages of the LSAD-CR-IDW method in local anomaly detection have not been highlighted, because the Segundo dataset is large, and its detection results are limited by the size of the detection window. In addition, compared with the real-time method RT-CR-RXD and PBS-FRT, the proposed FMLRT-RAD algorithm has almost the same detection results but is more prominent for anomaly targets.

In the Pavia Centre dataset, all anomalies of the FMLRT-RAD algorithm can be well detected, the shape and size are similar to ground truth, and real-time processing of the algorithm can be realized. Obviously, FEBPAD, VABS, CRDBPSW, and AHMID cannot detect some anomalies, while RGAE and local anomaly detection LSAD-CR-IDW methods detect all anomaly targets, but also misdetect some backgrounds as anomalies. RXD and RAD detect anomaly objects, but the anomaly shape is missing compared with the ground

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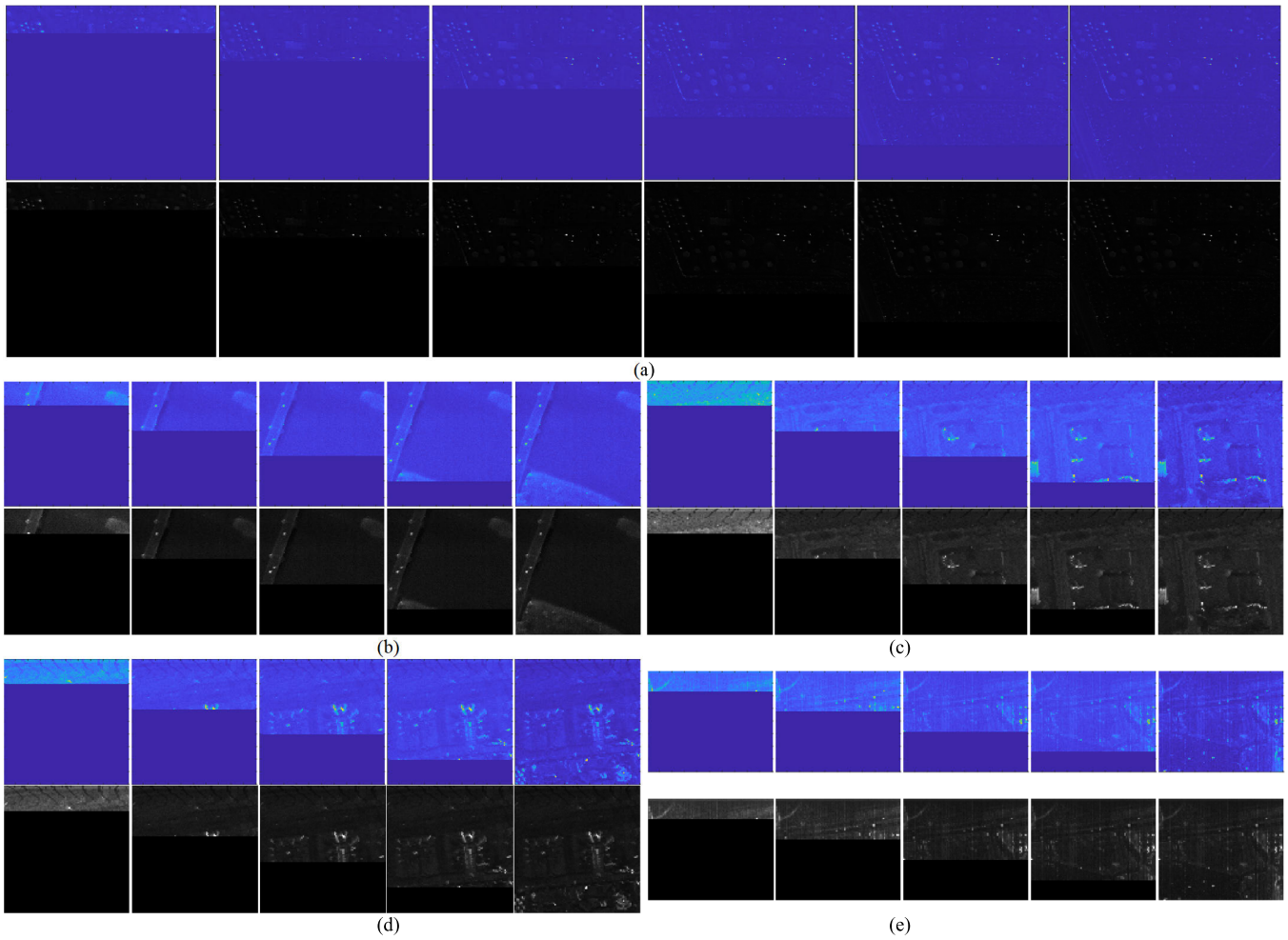


Fig. 13. Detection process of FMLRT-RAD on five experimental datasets. (a) Segundo. (b) Pavia Centre. (c) ABU. (d) Los Angeles. (e) HYDICE Urban.

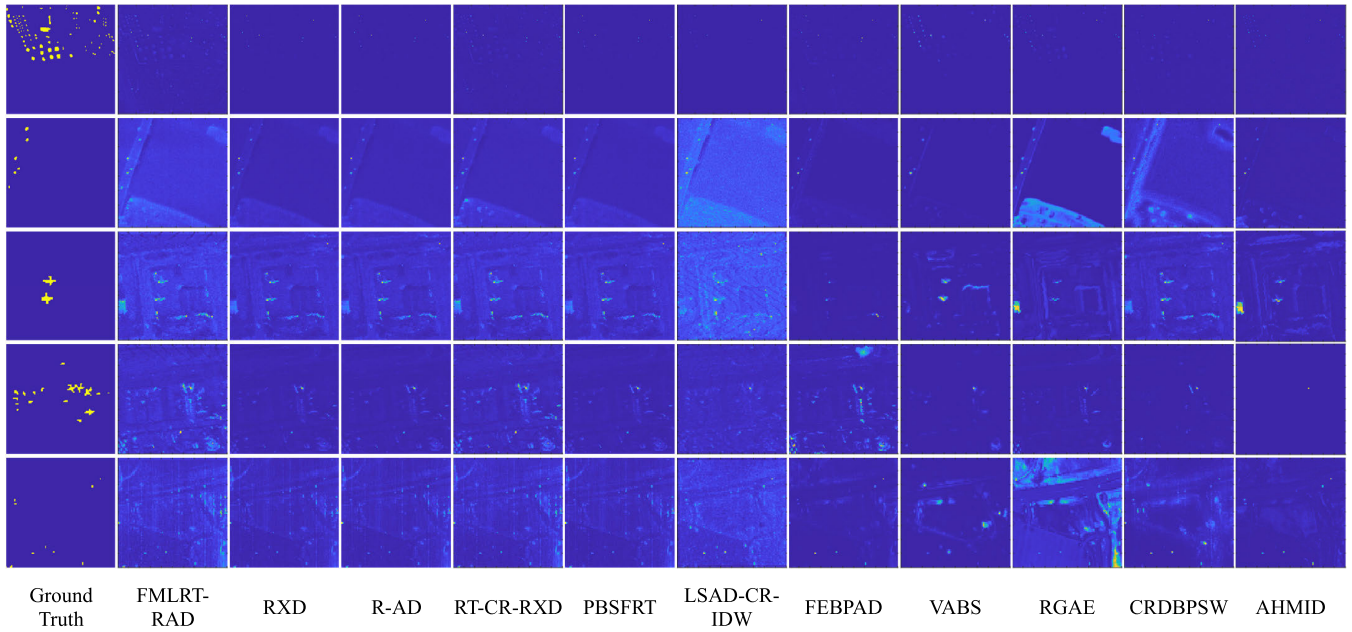


Fig. 14. Detection maps of different methods on five experimental datasets.

truth, and the real-time methods RT-CR-RXD and PBSFRT also have such problems.

In the ABU dataset, the proposed FMLRT-RAD algorithm tends to preserve the anomalies in the detection graph and can

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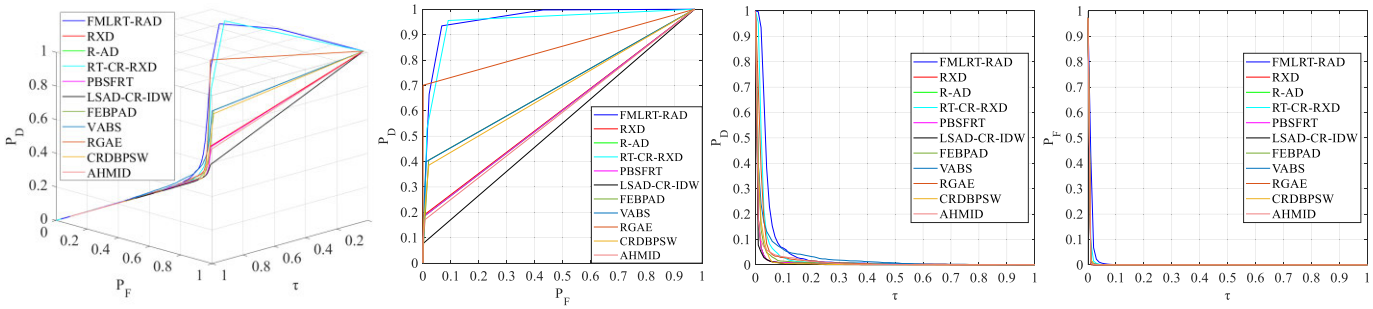


Fig. 15. ROC curves of different methods on Segundo. (a) 3-D ROC curves (P_D, P_F, τ). (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

realize real-time processing line-by-line. In contrast, FEBPAD hardly detected any anomalies, and the anomalies detected by RGAE are obviously different from the ground truth. The anomalies detected by VABS and AHMID are incomplete, and the nonanomaly targets are mistakenly detected as anomalies. In comparison, RXD, RAD, RT-CR-RXD, CRDBPSW, and PBSFRT have almost the same detection results as the proposed method, while RXD, RAD, and PBSFRT cannot achieve line-by-line real-time detection. In addition, the background suppression of the LSAD-CR-IDW method is much different from that of the FMLRT-RAD algorithm in local anomaly detection, which makes the anomaly drown in the background.

In the Los Angeles dataset, the FMLRT-RAD algorithm can clearly detect anomalies and can detect them in real-time after obtaining the data line-by-line. In contrast, AHMID detects almost no anomaly, while VABS, CRDBPSW, and RGAE detect more anomalies than AHMID, but still miss most of them. RXD, RAD, FEBPAD, and PBSFRT detected many partial anomalies but did not detect local anomalies, and even FEBPAD mistakenly detected the background as an anomaly. Local anomaly detection LSAD-CR-IDW and real-time RT-CR-RXD methods detected almost all anomalies, but the anomaly results of LSAD-CR-IDW were not obvious.

In the HYDICE Urban dataset, the FMLRT-RAD algorithm detects almost all anomalies and implements local real-time processing of BIL data acquisition. The detection result of the RGAE method is different from the ground truth. FEBPAD, VABS, CRDBPSW, and AHMID suppress the local anomalies although the background is well suppressed. For relatively weak local anomalies, RXD and RAD fail as global anomaly detection models. Real-time RT-CR-RXD and PBSFRT methods also have such problems. In local anomaly detection LSAD-CR-IDW, almost all anomaly targets are detected because of the use of local information.

It is worth noting that both the proposed FMLRT-RAD and RT-CR-RXD methods belong to line-by-line real-time anomaly detection. It is evident from Fig. 14 that the detection results of the two methods are similar across multiple datasets. However, it is interesting that the proposed FMLRT-RAD algorithm is more inclined to preserve the anomalies in the detection graph. By combining the line-by-line moving process in Fig. 4 and using changes in pixel information, we can find that different from the RT-CR-RXD method, which needs to record pixel information of all locations. In the proposed FMLRT-RAD algorithm, after obtaining the information on the current line and determining it as an anomaly line, the correlation matrix implements an update process that includes new line and excludes old ones. In other words, the FMLRT-RAD algorithm only needs to use part of the line information, which reduces the required storage space and helps to further realize the retention of anomaly data in local real-time anomaly detection.

In general, for data with weak or sparsely distributed anomalies, the global anomaly detection method will cause anomalies to drown in the background, while the local anomaly detection method has advantages over the global method, but is limited by the size of the detection window. In addition, although the advanced anomaly detection method can obtain better detection results, it cannot realize real-time processing of the data obtained line-by-line, and can only wait for all the data to be obtained. By comparing with nine comparison methods on five datasets, the proposed method is proved to be superior in anomaly detection performance for hyperspectral data of different sizes and categories.

B. 3-D ROC Analysis

In addition, to conduct a quantitative study of detection performance, Figs. 15–19 are drawn, showing the ROC curves of various methods on the five datasets. The AUC values of each method on five datasets are shown in Tables IV–VIII, where red is the best and green is the next best.

For the Segundo dataset, it can be seen from the 3-D ROC curve and the corresponding 2-D ROC curve in Fig. 15 that the FMLRT-RAD proposed in this article has the highest detection rate. Specifically, for the Segundo dataset, the 2-D ROC curve (P_D, P_F) of FMLRT-RAD is more inclined to the upper left, while the 2-D ROC curve (P_D, τ) is more inclined to the upper right, indicating that the detected anomaly targets have stronger retention ability, and the proposed algorithm has better detection performance. In addition, the AUC values for each method on the Segundo dataset are shown in Table IV, where red represents the best and green is the next best. It can be seen that the AUC (P_D, P_F) and AUC (P_D, τ) values of FMLRT-RAD are higher than those of the comparison method, which are 0.9364 and 0.0483, respectively. Although the AUC (P_F, τ) value of FMLRT-RAD is not the best, it is close to the

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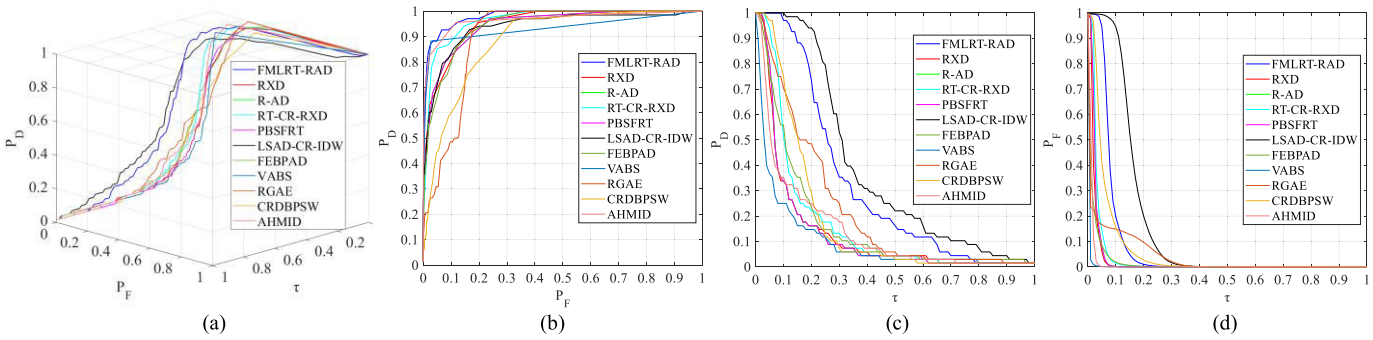


Fig. 16. ROC curves of different methods on Pavia Centre. (a) 3-D ROC curves (P_D, P_F, τ). (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

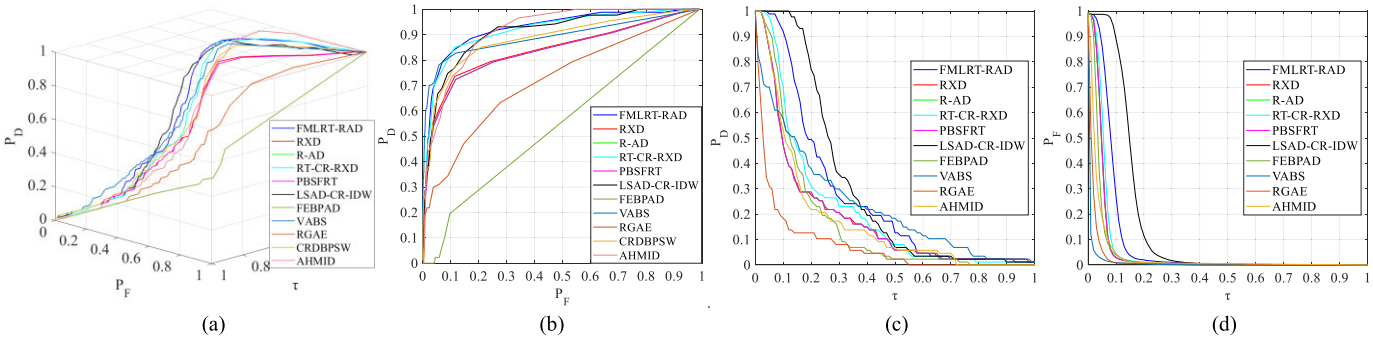


Fig. 17. ROC curves of different methods on ABU. (a) 3-D ROC curves (P_D, P_F, τ). (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

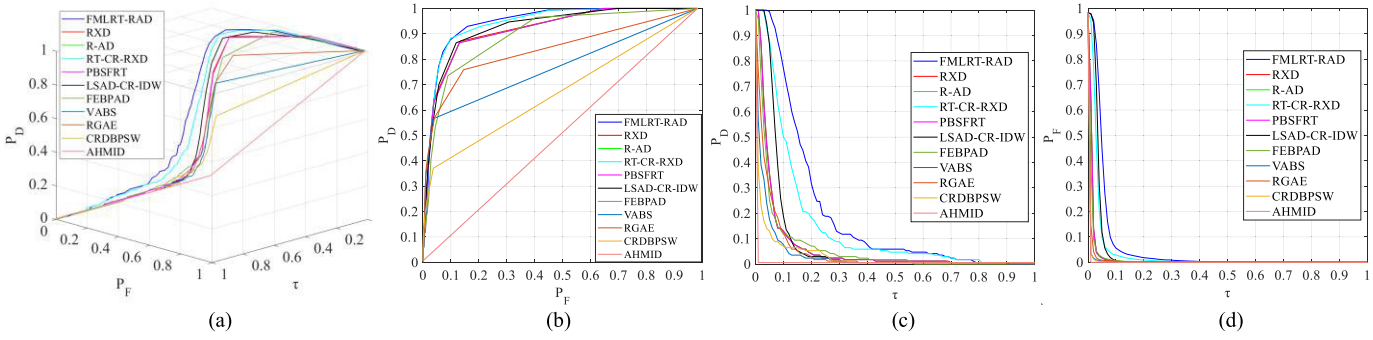


Fig. 18. ROC curves of different methods in Los Angeles. (a) 3-D ROC curves (P_D, P_F, τ). (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

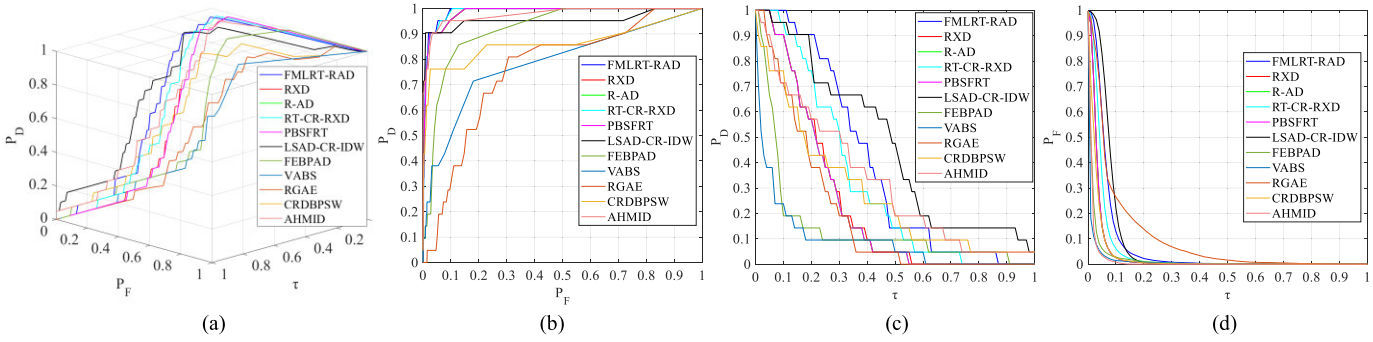


Fig. 19. ROC curves of different methods on HYDICE Urban. (a) 3-D ROC curves (P_D, P_F, τ). (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

best. The results show that FMLRT-RAD has a high detection rate.

The ROC curve of each method on the Pavia Centre dataset is shown in Fig. 16. As can be seen from the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 16, the comprehensive performance of FMLRT-RAD proposed in this article is very good. As shown in Table V, the AUC (P_D, P_F) value of the proposed FMLRT-RAD is significantly superior to the

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TABLE IX
AUC VALUES OF DIFFERENT MODULES OF PROPOSED METHOD ON PAVIA CENTRE

Method	AUC of (P_D, P_F)	AUC of (P_D, τ)	AUC of (P_F, τ)
FMLRT-RAD-L	0.9598	0.1961	0.0456
FMLRT-RAD-IOL	0.9605	0.1704	**0.0275**
FMLRT-RAD (FMLRT-RAD-IOL and FMLRT-RAD-L)	**0.9784**	**0.3193**	0.0829

other nine comparison methods, and the AUC (P_D, τ) value of FMLRT-RAD is second only to the optimal value of LSAD-CR-IDW. In addition, the AUC (P_F, τ) value of FMLRT-RAD is better than that of LSAD-CR-IDW, indicating that the proposed method has the best comprehensive performance.

The ROC curve of each method on the ABU dataset is shown in Fig. 17. It can be seen from the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 17 that the FMLRT-RAD proposed in this article has a better comprehensive performance. As shown in Table VI, the AUC (P_D, P_F) value of the proposed FMLRT-RAD is superior to other comparison methods. The AUC (P_D, τ) value of FMLRT-RAD is second only to the best value of LSAD-CR-IDW. In addition, the AUC (P_F, τ) value of FMLRT-RAD is better than that of LSAD-CR-IDW, indicating that the background compression effect is better than that of LSAD-CR-IDW.

The ROC curve of each method on the Los Angeles dataset is shown in Fig. 18. As can be seen from the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 18, the comprehensive performance of FMLRT-RAD proposed in this article is very good. Table VII shows that the AUC (P_D, P_F) value of the proposed FMLRT-RAD is better than that of the other nine comparison methods. At the same time, the AUC (P_D, τ) value of the proposed FMLRT-RAD obtains the best value of all comparison methods, indicating that the proposed method has a stronger retention ability for anomalies on the Los Angeles dataset. It is worth noting that FMLRT-RAD is superior to RT-CR-RXD in AUC (P_D, P_F) and AUC (P_D, τ) values, indicating that the proposed method has more obvious advantages for line-by-line anomaly detection.

The ROC curve of each method on the HYDICE Urban dataset is shown in Fig. 19. It can be seen from the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 19 that the proposed FMLRT-RAD has a better comprehensive performance. As shown in Table VIII, the AUC (P_D, P_F) value of the proposed FMLRT-RAD is significantly superior to other comparison methods, and the AUC (P_D, τ) value is second only to LSAD-CR-IDW. In addition, the AUC (P_F, τ) value of FMLRT-RAD is better than that of LSAD-CR-IDW, indicating that the background compression effect is better than that of LSAD-CR-IDW.

In addition, to compare the performance of each method more intuitively, Fig. 20 shows the comparison of ROC histograms of different methods on five datasets. To fully display the data names, the Segundo, Pavia Centre, ABU, Los Angeles, and HYDICE Urban are named D1, D2, D3, D4, and D5, respectively.

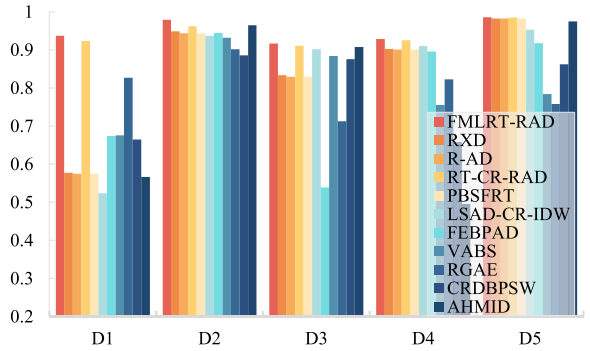


Fig. 20. Comparison of AUC (P_D, P_F) values of different methods on five datasets.

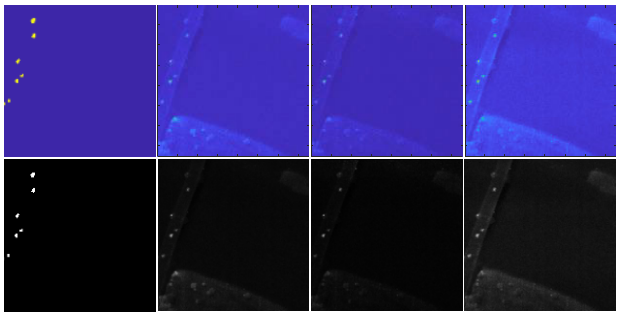


Fig. 21. Detection maps of different modules of proposed method on the Pavia Center.

According to the data analysis results in Fig. 20, it can be concluded that the FMLRT-RAD algorithm proposed in this article performs well on five real datasets, and its AUC (P_D, P_F) value is superior to all other comparison algorithms, indicating that this method has significant advantages in anomaly detection performance. Although the FMLRT-RAD algorithm may not be optimal in terms of AUC (P_D, τ) and AUC (P_F, τ) values, it needs to be considered that in some of the better target-retention methods, the detection rate may be reduced due to the retention of misdetected anomalies. However, the proposed FMLRT-RAD method successfully detects almost all anomalies and completes the reliable detection of these anomalies. In summary, the FMLRT-RAD algorithm proposed in this article is superior to other methods in the application of anomaly detection of real hyperspectral data. It performs well in real-time processing, providing a more reliable and efficient solution for a wide range of real-world real-time anomaly detection applications.

C. Ablation Experiments

FMLRT-RAD incorporates two newly added modules. To highlight the contributions of these two modules to anomaly detection, we separately employ each module as a method for anomaly detection. Subsequently, we compare each individual module's performance with the proposed method, which combines both modules, as shown in Fig. 21.

From Fig. 21, it can be observed that the method employing only the FMLRT-RAD-IOL module provides clearer detection for certain anomalies compared to the FMLRT-RAD-L

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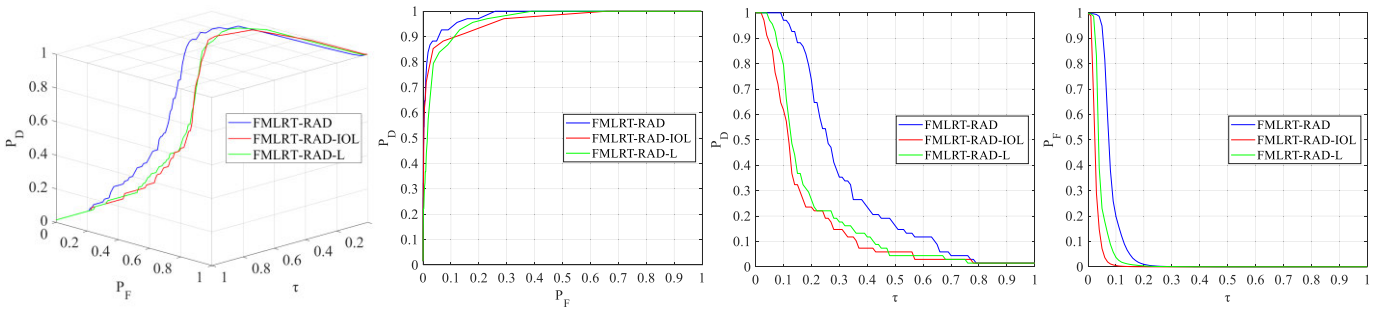


Fig. 22. ROC curves of different methods on Pavia Centre. (a) 3-D ROC curves (P_D, P_F, τ). (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

method. In comparison to methods using each module separately, the proposed approach, which combines both modules, achieves more accurate detection of anomalies of different sizes.

From Table IX and Fig. 22, it can be observed that the values of (P_D, P_F) and (P_D, τ), for the method using only the FMLRT-RAD-IOL module and the FMLRT-RAD-L module are both lower than those for FMLRT-RAD. Specifically, the (P_D, P_F) value for the FMLRT-RAD-IOL module is higher than that for the FMLRT-RAD-L method, indicating a more significant contribution of the FMLRT-RAD-IOL module to anomaly detection in the proposed approach FMLRT-RAD. However, the (P_D, τ) value for the FMLRT-RAD-L module is higher than that for the FMLRT-RAD-IOL method, suggesting a more pronounced contribution of the FMLRT-RAD-L module to anomaly preservation in the proposed approach. Additionally, in terms of background suppression, the FMLRT-RAD-IOL module exhibits an advantage over FMLRT-RAD-L of the values of (P_F, τ). Overall, the FMLRT-RAD-IOL module makes a more substantial contribution to the proposed approach, and the effectiveness of using the two modules together surpasses that of using them individually.

VI. Conclusion

The detection effect of the local anomaly detection method is limited by the size of the selected neighborhood window. However, due to the different requirements of different data, the fixed size window is difficult to meet the detection application of real-time data acquisition. To solve this problem, based on the most widely used push-broom hyperspectral imaging sensor to obtain data in a line-by-line type, this article proposes a line-by-line real-time local anomaly detection method based on the finite Markov model. By deeply analyzing the information contained in the data obtained line-by-line, the current line is divided into anomaly lines or nonanomaly lines and the corresponding correlation matrix calculation method is designed. Then, the updated equation is derived suitable for real-time local anomaly detection. For anomaly lines, we only need to use partial lines information, thereby reducing the need for full data. This approach avoids the interference information and irrelevant background contained in the entire data, which helps to preserve anomalies in local real-time anomaly detection and improve detection accuracy. The experimental results show that the algorithm performs well on multiple datasets, not only significantly reducing the statistical calculation requirements for all data, but also achieving real-time data processing. This fully shows that the algorithm has a wide range of application potential. It is worth noting that although research focuses on the push-broom hyperspectral imaging sensor, conclusions can be generalized to other hyperspectral image acquisition methods that require real-time processing, such as BSQ format. For real-time hyperspectral anomaly detection technology, the algorithm's computational cost is a crucial performance indicator. In recent years, there has been widespread attention to techniques based on GPU parallel processing. Therefore, the next research interest and direction will focus on further utilizing GPU parallel processing methods to accelerate the speed. In addition, during hyperspectral imaging, the data is often subject to various degradation, noise effects, or variability [50]. Therefore, addressing variability in images will be another key aspect of our future work.

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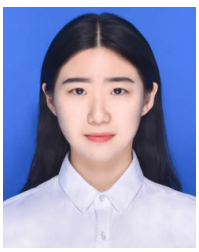
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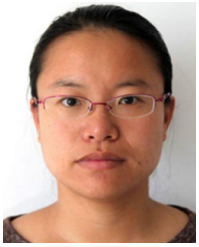
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Shihui Liu is currently pursuing the Ph.D. degree in computer science and technology with Dalian Maritime University, Dalian, China.

Her research interests include remote sensing and hyperspectral image processing.

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Meiping Song (Member, IEEE) received the Ph.D. degree from the School of Computer Science and Technology, Harbin Engineering University, Harbin, China, in 2006.

From 2013 to 2014, she was a Visiting Assistant Researcher with the University of Maryland, College Park, MD, USA. She is currently a Professor with the College of Information Science and Technology, Dalian Maritime University, Dalian, China. Her research interests include remote sensing and hyperspectral image processing.



Bing Xue (Fellow, IEEE) received the B.Sc. degree from the Henan University of Economics and Law, Zhengzhou, China, in 2007, the M.Sc. degree in management from Shenzhen University, Shenzhen, China, in 2010, and the Ph.D. degree in computer science from the Victoria University of Wellington (VUW), Wellington, New Zealand, in 2014.

She is currently a Professor of artificial intelligence and the Deputy Head of the School of Engineering and Computer Science, VUW, where she is the Deputy Director of the Center for Data Science and Artificial Intelligence. She has over 400 papers published in fully refereed international journals and conferences. Her research focuses mainly on evolutionary computation, machine learning, classification, symbolic regression, feature selection, evolving deep neural networks (NNs), image analysis, transfer learning, and multiobjective machine learning.

Dr. Xue is a fellow of the Engineering New Zealand. She has been organizing many international conferences, such as the General Chair of Pacific Rim International Conference on Artificial Intelligence (PRICAI) 2025, the Conference Chair of IEEE Congress on Evolutionary Computation (CEC) 2024, and EuroGP 2024. She has also served as an Associate Editor for several international journals, such as IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION (TEVC), IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE (TAI), *IEEE Computational Intelligence Magazine* (CIM), and ACM TRANSACTIONS ON EVOLUTIONARY LEARNING AND OPTIMIZATION (TELO).



Chein-I Chang (Life Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

Since 2016, he has been a Chang Jiang Scholar Chair Professor and the Director of the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Dalian Maritime University, Dalian, China, and a Yushan Scholar Chair Professor with National Cheng Kung University, Tainan, Taiwan. In addition, he has been a Chair Professor with National Yang Ming Chiao Tung University, Hsinchu, Taiwan, since 2019. He is currently a Professor with the University of Maryland Baltimore County (UMBC), Baltimore, MD, USA. He has authored four books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic Publishers, 2003), *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection* (Springer, 2016), and *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017). He also edited three books, *Recent Advances in Hyperspectral Signal and Image Processing* (2006), *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007), and *Advances in Hyperspectral Image Processing Techniques* (Wiley, 2023). In addition, he also co-edited with A. Plaza a book on *High Performance Computing in Remote Sensing* (CRC Press, 2007). His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of the Society for Photo-Optical Instrumentation Engineers (SPIE). He is currently serving as an Associate Editor for *Remote Sensing* and IEEE TRANSACTION ON GEOSCIENCE AND REMOTE SENSING.



Mengjie Zhang (Fellow, IEEE) received the B.E. and M.E. degrees from the Artificial Intelligence Research Center, Agricultural University of Hebei, Hebei, China, in 1989 and 1992, respectively, and the Ph.D. degree in computer science from RMIT University, Melbourne, VIC, Australia, in 2000.

He is currently a Professor of computer science and the Director of the Centre for Data Science and Artificial Intelligence, Victoria University of Wellington, Wellington, New Zealand. He has published over 900 research papers in refereed international journals and conferences. His research interests include genetic programming, image analysis, feature selection and reduction, job-shop scheduling, and evolutionary deep learning and transfer learning.

Dr. Zhang is a fellow of the Royal Society of New Zealand, a fellow of the Engineering New Zealand, and an IEEE Distinguished Lecturer.

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